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From 6D $(2,0)$ Theory to Instanton Observables: DAHA, Theta Kernels, and BPS Structures

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Abstract: We explore possible connections between double affine Hecke algebras (DAHA), the geometry of ADE singularities in six-dimensional $(2,0)$ theory, and the four-dimensional Ω-background. We study the dual roles of the X- and Y-variables, the degeneration of DAHA kernels to Weyl denominators, and their relation to theta functions, curvature, and modular geometry. We further examine periodic skew Young diagrams as a combinatorial description of DAHA representations and investigate their possible relation to Nekrasov instanton observables and qq-characters. Finally, we propose a non-Archimedean perspective on the (q,t)-deformation and formulate conjectural connections with BPS filtrations. These constructions provide an algebraic framework for investigating how singularity geometry, representation theory, and protected quantum sectors may be related.

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In $6D$ $(2,0)$ theory with 16 supercharges, it is well known that no Lagrangian description is available. Instead, the theory is encoded purely in geometric data—BPS conditions(chiral), curvature, and ADE singularities—making it feel less like a conventional field theory and more like a carefully arranged piece of geometry that refuses to be written in coordinates.

Its superconformal symmetry is $OSp(8^*|4)$, whose bosonic subgroup is $SO(6,2) \times USp(4)_R \cong SO(6,2) \times SO(5)_R$. The supercharges transform as two parts: on one side, usually one chooses a single chirality of the $6D$ Lorentz group $SO(5,1)$, namely the positive-chirality Weyl spinor 4^+, with spinor index $\alpha = 1, \cdots, 4$. On the R-symmetry side, $Spin(5) \cong Sp(2)$ admits a quaternionic fundamental representation, which is also 4-dimensional; we label its indices by $a = 1, \cdots, 4$. Combining these two factors, the supercharges carry the structure Q_a^α, giving $4 \times 4 = 16$ components in total. These are subject

to a real symplectic Majorana–Weyl condition, ensuring the correct reality structure for 6D $(2,0)$ supersymmetry.

At first glance, it looks like there are $6 \times 5 = 30$ independent components. However, this object is a mixed-index tensor, and supersymmetry together with Lorentz and $SO(5)$ R-symmetry imposes strong constraints on its structure. After taking these constraints into account, the 30 components reorganize into physically meaningful sectors: 6 components corresponding to momenta, 10 components associated with self-dual string charges, and 5 components transforming as scalar (five-brane) central charges under $SO(5)$ We denote them as follows

$$P^{\mu}(6) \quad H_{\mu\nu\rho} = +\frac{1}{3!}\varepsilon_{\mu\nu\rho\sigma\lambda\tau}H^{\sigma\lambda\tau}(10) \quad Z^{I}(5) \tag{0.1}$$

where

$$H_{\mu\nu\rho} = 3\partial_{[\mu}B_{\nu\rho]} \tag{0.2}$$

and gauge field $B_{\mu\nu}$ transforms as

$$B_{\mu\nu} \to B_{\mu\nu} + \partial_{\mu}\Lambda_{\nu} - \partial_{\nu}\Lambda_{\mu} \tag{0.3}$$

Since there is no conventional gauge field in the theory, the natural charged objects are self-dual strings. Their charges are carried by a vector e^{I}_{μ}, where μ labels spacetime directions and I labels the transverse scalar (or R-symmetry) directions. These objects can be understood as arising from $M2$-branes ending on $M5$-branes.

Since the theory is governed purely by geometric constraints, the BPS condition can be viewed as a kind of chiral restriction. In particular, near a curve (a self-dual string) associated with an ADE -type singularity, it is natural to introduce two operations: one corresponding to a loop (or rotation) around the singularity(give a parameter q), and another corresponding to a reflection in the root lattice, relating it to neighboring singularities(give a parameter t). By adding degenerate or stable limiting objects, the moduli space admits a natural compactification, and the resulting curves become compactified in the sense of Deligne–Mumford compactification, already hinting at a categorified structure. In this framework, we use the weight and coweight lattices to encode the operators introduced earlier. Two independent periodicities then emerge: one associated with loops around the singularities, and another arising from the affine of the Lie algebra which encoded singularities information. Together, these combine to produce an effectively double torus-like structure.

The purpose of this work is to explore the geometric structures that may connect the six-dimensional $(2,0)$ theory, double affine Hecke algebras (DAHA), and four-dimensional instanton observables. Rather than assuming a direct equivalence between these frameworks, we examine several concrete intermediate structures, including theta-function kernels, Weyl denominators, Dunkl–Cherednik operators, and periodic skew Young diagrams. We distinguish the relations obtained from explicit calculations from the conjectural interpretations involving six-dimensional geometry and BPS filtrations.

When reformulated in analytic terms via Hilbert schemes and affine Grassmannians, the algebraic structure admits a clear geometric interpretation: the parameter q can be

understood as the loop parameter, while t plays the role of a grading variable, often interpreted as a cotangent or equivariant grading. From this perspective, the t-grading naturally lives in cohomology or K-theory, encoding the underlying geometric refinement of the construction.

One may confuse about the curvature information. Before addressing this question, we first introduce a powerful algebraic tool from the representation theory of quantum groups, Double Affine Hecke Algebra(DAHA), which allows us to reformulate the above discussion about singularities in a more structured way. We will then return to the issue of curvature in a later subsection.

1 DAHA as deformation of singularities

The definition of double affine Hecke algebra (DAHA) is as following, (This formulation differs slightly from the definition given in the Appendix, where the algebra is introduced in its original and more general form. The present approach is more convenient for our purposes and makes the underlying structure more transparent. Nevertheless, it is important to keep in mind that the X-variables should be regarded as the fundamental generators of the theory. The Y-variables arise as derived objects, obtained from the action of the affine Hecke algebra on the X-lattice. In the simply-laced cases of type A, D, E, the two families are related by the DAHA Fourier transform, which exchanges the roles of X and Y. From this perspective, the Y-variables may be viewed as a mirror image of the X-variables, reflecting the intrinsic duality, which is Weyl group, built into the double affine structure.

Definition 1.1. The double affine Hecke algebra $\mathcal{H}_{q,t}$ is generated over the field $\mathbb{Q}_{q,t}$ by the elements $\{T_i, 0 \leq i \leq n\}$ and two sets pairwise commutative $\{X_b, b \in P\}$ and $\{Y_b, b \in P\}$, where $b = \sum_{i=1}^{n} l_i \omega_i$, and Weyl group Π, where following relations are imposed,

- (i) $(T_i - t^{1/2})(T_i + t^{-1/2}) = 0 \quad 0 \leq i \leq n$
- (ii) $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1} \quad i < n-1, j > i+1$
- (iii) $X_{b_1} X_{b_2} = X_{b_2} X_{b_1}$,$Y_{b_1} Y_{b_2} = Y_{b_2} Y_{b_1}$, where $b_1, b_2 \in P$
- (iv) $T_i X_b - X_{s_i(b)} T_i = (t_i^{1/2} - t_i^{-1/2}) \frac{X_b - X_{s_i(b)}}{1 - X_{-\alpha_i}}$, where α_i is the simple roots.
- (v) $T_i Y_b - Y_{s_i(b)} T_i = (t_i^{1/2} - t_i^{-1/2}) \frac{Y_b - Y_{s_i(b)}}{1 - Y_{-\alpha_i}}$, where α_i is the simple roots.
- (vi) $X_{b_1} X_{b_2} = X_{b_2} X_{b_1}$,$Y_{b_1} Y_{b_2} = Y_{b_2} Y_{b_1}$, where $b_1, b_2 \in P$
- (vii) $T_i X_b T_i = X_b X_{\alpha_i}^{-1}$,$T_i^{-1} Y_b T_i^{-1} = Y_b Y_{\alpha_i}^{-1}$ where $(b, \alpha_i) = 0, 1 \leq i \leq n$
- (viii) $T_i X_b = X_b T_i$, $T_i Y_b = Y_b T_i$ if $(b, \alpha_i) = 0$
- (ix) $Y_b X_b Y_b X_b = T^2 q^{-1/2}$

Following the discussion above and the material reviewed in the Appendix, the appearance of the DAHA may be understood as a natural two-parameter ((q,t)) deformation associated with ADE-type singularities. From this perspective, DAHA encodes the algebraic structure governing the corresponding quantum geometry, where the parameters q and t measure the deformation away from the classical singular limit. In geometric realizations arising from brane constructions, the algebra naturally acts on the moduli of $D0$-brane configurations, and the variables X and Y may be interpreted as dual coordinates describing the resulting quantum dynamics (We would interpret X-varibles as instantons in 6D, Y-varibles as local Hamiltonian if there have an instanton.). This viewpoint suggests that DAHA should be regarded not merely as an algebraic deformation of Weyl-group symmetries, but as a structure capturing the interplay between ADE geometry, representation theory, and the dynamics of point-like branes. DAHA acts on

$$\mathbb{C}[T] \ltimes \mathbb{C}W \ltimes \mathbb{C}[T] \tag{1.1}$$

where W is the Weyl group of the singularity types, one torus the variables X lives on it, and another Y lives on it. Every element of DAHA can be uniquely written as

$$H = \sum_{w \in W} c_{a,w,b} X_a T_w Y_b \tag{1.2}$$

2 Generalized state-operator correspondence

Although a global Lagrangian description is generally unavailable in the presence of ADE singularities, one may nevertheless define local Hamiltonians in a neighborhood of each singular point. Since these operators probe only the local geometry, we propose that, after fixing the singularity data, they form a mutually commuting family,

$$[Y_a, Y_b] = 0, \qquad a, b \in P, \tag{2.1}$$

where P denotes the corresponding weight lattice. From the DAHA perspective, these local Hamiltonians generate the commutative Y-subalgebra, while the affine Weyl group controls their transformation properties. Thus the local dynamics is encoded not by a single global Hamiltonian, but rather by a collection of commuting Hamiltonians attached to the singular strata of the geometry. In the polynomial representation, these operators become the Macdonald-type difference operators, whose simultaneous eigenstates provide a natural basis for the local quantum theory.

The dual variables X_a describe the corresponding instanton sectors and satisfy

$$X_a X_b = X_{a+b}, \qquad Y_a Y_b = Y_{a+b} \tag{2.2}$$

By the PBW decomposition,

$$\mathcal{H} \simeq \mathbb{C}[X] \otimes H_W(t) \otimes \mathbb{C}[Y], \tag{2.3}$$

the X- and Y-sectors appear on an equal footing, and the affine Weyl group intertwines the two descriptions. We therefore interpret the correspondence between X and Y as

a twisted state–operator correspondence: instanton configurations represented by the X-variables are paired with local Hamiltonians represented by the Y-variables. It should be emphasized, however, that the Hamiltonians are not independent observables. Rather, they acquire physical significance only when the corresponding instanton sector is present. Schematically, one may view this relation as

$$X_b \longleftrightarrow Y_b, \tag{2.4}$$

where the existence of an instanton state X_b determines the geometric background on which the local Hamiltonian Y_a acts. In this sense, the instanton sector provides the support of the local dynamics, while the Hamiltonian sector measures its quantum fluctuations.

3 From DAHA Kernels to Curvature

In this section, we study the geometric meaning of the DAHA kernel through its relation to Jacobi theta functions and the Weyl denominator. We show how the (q)-deformed kernel degenerates to the classical Weyl denominator and interpret the theta-function ratio in terms of Hermitian metrics, curvature, and divisor geometry. This provides a geometric perspective connecting the elliptic and classical settings.

It has been shown that the classical Fourier transform is recovered from the kernel

$$\delta_k(x;q) = \prod_{j=0}^{\infty} \frac{(1-q^{j+2x})(1-q^{j-2x})}{(1-q^{j+k+2x})(1-q^{j+k-2x})}, \quad 0<q<1, k\in\mathbb{C} \tag{3.1}$$

In Macdonald theory and rational DAHA ($t_\alpha = q_\alpha^{k_\alpha}$), for a reduced root system $\triangle$, it is defined as [2]

$$\delta(X;q,t) = \prod_{\alpha\in\triangle^+}\prod_{i=0}^{\infty} \frac{(1-X_\alpha q_\alpha^i)(1-X_\alpha^{-1}q_\alpha^i)}{(1-X_\alpha t_\alpha q_\alpha^i)(1-X_\alpha^{-1}t_\alpha q_\alpha^i)} \tag{3.2}$$

For instance, for A_1 type, by Pochhammer symbols($(a;q)_\infty = \prod_{i=0}^{\infty}(1-aq^i)$), it looks like

$$\delta(X;q,t) = \prod_{i=0}^{\infty} \frac{(1-Xq^i)(1-X^{-1}q^i)}{(1-Xtq^i)(1-X^{-1}tq^i)} = \frac{(X;q)_\infty (X^{-1};q)_\infty}{(tX;q)_\infty (tX^{-1};q)_\infty} \tag{3.3}$$

Recall the Jacobi theta function

$$\theta(X,q) = (X;q)_\infty (qX^{-1};q)_\infty (q;q)_\infty \tag{3.4}$$

thus for A_1 case

$$\delta(X;q,t) = \frac{\theta(X,q)}{\theta(X,q,t)} \tag{3.5}$$

Thus, in the DAHA framework, the kernel is the "fake" q-adic theta functions. Using condition $t=q^k$ to simplify, we would get

$$\frac{(X;q)_\infty}{(q^kX;q)_\infty} = \prod_{j=0}^{k-1}(1+Xq^j) \tag{3.6}$$

thus

$$\delta(X;q,t)=\prod_{j=0}^{k-1}(1-Xq^j)(1-X^{-1}q^j) \tag{3.7}$$

For reduced root systems,

$$\delta(X;q,t)=\prod_{\alpha\in\triangle^+}\prod_{j=0}^{k_\alpha-1}(1-X_\alpha q_\alpha^j)(1-X_\alpha^{-1}q_\alpha^j) \tag{3.8}$$

We do variables changes to make it more familiar, set

$$X_\alpha=e^{\epsilon\alpha(x)}\quad q=e^{-\epsilon}\quad t=e^{-k\epsilon} \tag{3.9}$$

then

$$\delta(X;t,q)=\lim_{\epsilon\to 0}\prod_{\alpha\in\triangle^+}\prod_{j=0}^{k_\alpha-1}(1-X_\alpha q_\alpha^j)(1-X_\alpha^{-1}q_\alpha^j) \tag{3.10}$$

$$=\prod_{\alpha\in\triangle^+}(e^{\alpha(x)/2}-e^{-\alpha(x)/2})^{2k_\alpha}=\prod_{\alpha\in\triangle^+}(2\sinh(\alpha(x)/2))^{2k_\alpha} \tag{3.11}$$

we come back to Weyl denominator.

Recall that the Jacobi theta function behaves like a section for a line bundle over an elliptic curve. The logarithm of its norm produces a natural potential, and taking derivatives of this potential gives the curvature of the corresponding bundle. On line bundle

$$|\theta(x+\tau,\tau)|^2=e^{-\pi i\tau-2\pi iz}|\theta(x,\tau)|^2 \tag{3.12}$$

could define a local Hermitian metric;

$$h(x)=e^{i\pi\tau}|\theta(x,\tau)|^2 \tag{3.13}$$

the curvature is

$$F=\frac{2\pi i}{\tau}dx\wedge d\bar{x} \tag{3.14}$$

In our case, we take

$$h(x)=e^{i\pi\tau}|\frac{\theta(X,q)}{\theta(X,q,t)}|^2 \tag{3.15}$$

then

$$F=-\partial\bar{\partial}h=-h\left(\partial\log\theta(X;q)-\partial\log\theta(X;q,t)\right)\wedge\left(\bar{\partial}\log\overline{\theta(X;q)}-\bar{\partial}\log\overline{\theta(X;q,t)}\right). \tag{3.16}$$

We use notation

$$F(X)=\frac{\theta(X;q)}{\theta(X;q,t)}. \tag{3.17}$$

To study the curvature, we consider the logarithm of h,

$$\log h=i\pi\tau+\log|F|^2. \tag{3.18}$$

Since the factor $e^{i\pi\tau}$ contributes only a constant term, it follows that

$$\partial\bar{\partial}\log h = \partial\bar{\partial}\log|F|^2. \tag{3.19}$$

The Poincaré–Lelong formula then implies

$$\partial\bar{\partial}\log|F|^2 = 2\pi i\left(\mathrm{Div}(F) - \mathrm{Div}_\infty(F)\right), \tag{3.20}$$

where $\mathrm{Div}(F)$ and $\mathrm{Div}_\infty(F)$ denote the zero and pole divisors of F, respectively. Consequently, the curvature form is supported precisely on the zeros and poles of the theta-function ratio $F(X) = \theta(X;q)/\theta(X;q,t)$. In geometric terms, the logarithmic Laplacian of the theta ratio measures the difference between the zero divisor of $\theta(X;q)$ and the zero divisor of $\theta(X;q,t)$, thus obtain the first Chern form of the associated line bundle. Such expressions occur frequently in the construction of metrics, theta line bundles, automorphic forms, and the geometric representation theory of affine and double affine Hecke algebras.

4 Does there are instantons in 6D?

From a standard but nontrivial topological fact, one has

$$H^3(M_6,\mathbb{Z}) \tag{4.1}$$

,If the elliptic structure is imposed, it leads to theta-function–like structures, as suggested by our previous analysis. Indeed, for self-dual fields in six dimensions, the partition function is typically realized as a section over the intermediate Jacobia

$$Jac(M_6) = H^3(M_6,\mathbb{R})/H^3(M_6,\mathbb{Z}) \tag{4.2}$$

so that the resulting object behaves like a vector-valued theta function rather than an ordinary scalar partition function.

In M-theory, the 6D $(2,0)$ theory arises from stacks of M5-branes, while M2-branes can stretch between sheets or defects. These configurations descend in six dimensions to self-dual strings. If one interprets singularities as punctures, defects, or branch points, they organize into string webs or trajectories; yet in the effective 6D description they still behave, in a coarse sense, like point-like singular loci. In lower-dimensional semiclassical limits (notably 5D and 4D reductions), such singular loci are connected by finite WKB trajectories, whose enumeration contributes to protected sectors of the partition function. Analogously, in six dimensions—after passing to a section over the intermediate Jacobian and thus isolating a single geometric sector—we may interpret nonperturbative contributions as arising from sums over finite WKB-type trajectories connecting singularities in the 6D theory. Jokingly, one can think of this as "instantons living between singularities"—a somewhat playful way to describe the nonperturbative trajectories connecting defect loci.

From the discussion above, if one considers k instantons, they can be viewed as equivalent in the sense that the deformation parameter t governs their transitions between different singular loci. In this picture, the configurations organize into a kind of crystallized

basis. Since we are in the affine setting, the weights of $U_q(g)(g = su(n), so(2n), E_6, E_7, E_8,$ namely ADE)can be encoded by charged Young diagrams. The periodicity is controlled by the affine root α_0 (as in the appendix), so that each periodic skew diagram corresponds to a weight in the DAHA/Fock space description of instantons associated to singularities. In this framework, the braid/Hecke generators T_i (interpreted as winding around singularities) increase the content. The difference operators Y_i(viewed as local Hamiltonians) implement affine shifts. Consequently, a periodic skew diagram can be understood as an orbit under repeated Y-shifts, encoding the affine structure of the system. So from this perspective, "counting" should be understood as counting elements of the crystallized basis in this setting.

Inspried by the Heckman-Opdam operators, and we turn DAHA to rational DAHA. we choose parameterization

$$q = e^{-\epsilon} \quad t = e^{-\epsilon k} = q^k \tag{4.3}$$

if

$$X_i = e^{\epsilon x_i} \tag{4.4}$$

then

$$\tau_q f(X_i) = f(qX_i) = f(e^{\epsilon(x-1)}) = f(e^{\epsilon x}) - \epsilon \partial_{x_i} f(e^{\epsilon x}) \tag{4.5}$$

then the q-shift operators turns into differential operator

$$\tau_q = 1 - \epsilon \partial_x + \mathcal{O}(\epsilon^2) \tag{4.6}$$

In the case of DAHAs of type A_n, for singular k and generic q(which parameters satisfy $q, t = q^k$), the semisimple irreducible representations can be represented by periodic skew young diagrams using the technique of interwiners. Here the counting not depending on the usual Young diagram, but the Periodic skew diagram. We now suppose q is not a root of unity, and $q^i t^j, i, j \in \mathbb{Z}$ also do not represent nontrivial roots of unity. Let $t^r = q^s$, for integral r, s, and $r \neq 0, s > 0, (r, s) = 1$, setting rational $k = s/r$

Definition 4.1. [1] An infinite skew diagram δ without empty rows is called r-periodic of degree n if

1. it is invariant with respect to translation by a vector $w = (v, v - r)$ in the (i, j)-plane for a proper positive integer v

2. the subdiagram of δ with the rows in the range $1 \leq i \leq v$ contains exactly n boxes

the degree n of a s-partition of a skew diagram δ means that it represent a disjoint union of s skew subdigrams $\{\delta_0, \cdots, \delta_{s-1}\}$, and the portions of this diagram δ creates consecutive paths from a certain system of skew paths without crossings. Thr different paths from such a system may have conincidance subpaths but they do not cross with each other. We count these the paths from bottom to top.

We number the boxed in skew Young diagram in the inverse order, from the last box of the last row through the first box of the first box of the first row, set

$$\xi_l = c_{p_l} + k(i_l - j_l), \quad l \in 1, \cdots, n \tag{4.7}$$

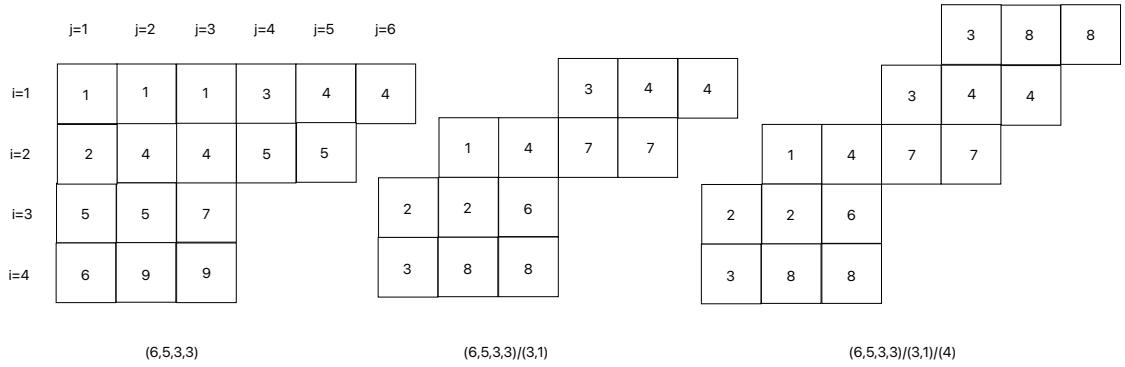


Figure 1. Young diagrams in standard labels

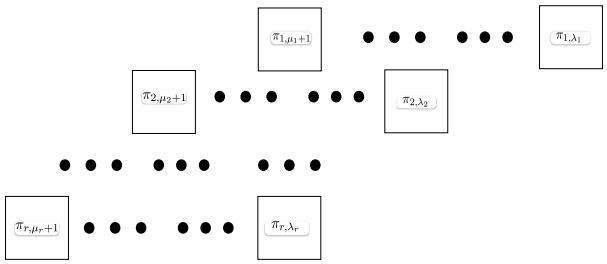


Figure 2. Skew Young diagram example

since there may have central elements, we denote them as c. If a perticular c assigned to periodic δ then the c number for the subdiagram $\tilde{\delta}$ we denote $c+(i-j)k$(here the (i,j) should module out of the peridoic by the imginary root, for instance, for $A_n^{(1)}$, $(i-j)(mod\, n)$, where $\alpha_0 = \alpha_1 + \cdots + \alpha_n$) for the coordinates (i,j) of the first box in the first nonempty row of the subdiagram $\tilde{\delta}$.

Let $\square$ be a set of skew Yonung diagram without empty rows of the total degree n, $C = \{c_1, \cdots, c_m\}$ be a sset of complex numbers such that $q^{c_a - c_b} \notin \mathbb{Z}(t,q), a \neq b$.

Theorem 4.1. *[2] An arbitrary irreducible X-semisimple $\mathfrak{H}$ can be constructed by a proper pair $\{\square, C\}$. Isomorphic modules correspond to the pairs which can be obtained from each other by a permutation of the skew diagrams in $\square$ together wiht the corresponding numbers in C, and by adding arbitrary integers to the c-numbers.*

A s-partition of $\{\square, C\}$ is a collection, namely $\{\tilde{\delta}_a^p | 0 \leq a \leq s, 1 \leq p \leq m\}$ of increasing skew partitions $\tilde{\delta}^p = \cup_a \tilde{\delta}_a^p$ of some fundamental subdiagrams $\tilde{\delta}^p$ of δ^p. Two pairs $\{\square, C\}$ be called SL-equivalent if the corresponding weights can be obtained from each other by adding a common constant.

Suppose that $\lambda = (\lambda_1, \lambda_2, \cdots, \lambda_r)$ and $\mu = (\mu_1, \mu_2, \cdots, \mu_r)$ be $r-$tuples of integers and

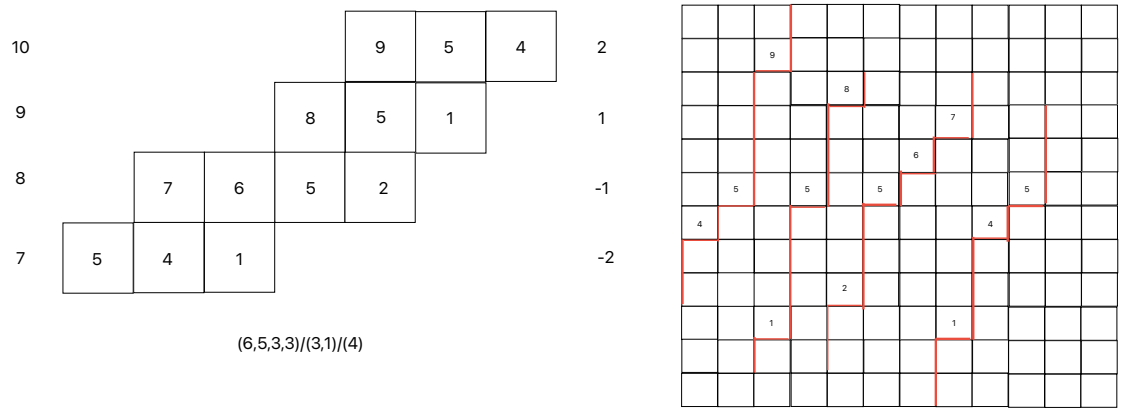


Figure 3. Skew Young diagram and Paths in $\mathbb{Z}^2$

let d be an integer such that

$$\lambda_1 \geq \lambda_2 - 1 \geq \lambda_3 - 2 \geq \cdots \geq \lambda_r - (r-1) \geq \lambda_1 - d - r \tag{4.8}$$

$$\mu_1 \geq \mu_2 - 1 \geq \mu_3 - 2 \geq \cdots \geq \mu_r - (r-1) \geq \mu_1 - d - r \tag{4.9}$$

$$\tag{4.10}$$

and let $a_1, a_2, \cdots, a_r$ and $b_1, b_2, \cdots, b_r$ be integers such that

$$a_1 - 1 \geq a_2 - 2 \geq a_3 - 3 \geq \cdots \geq a_r - r \geq a_1 - 1 - r \tag{4.11}$$

$$b_1 - 1 \geq b_2 - 2 \geq b_3 - 3 \geq \cdots \geq b_r - r \geq b_1 - 1 - r \tag{4.12}$$

$$\tag{4.13}$$

in [6], they proofed that cylindric partitions of shape $\lambda/\mu/d$ in whihc the entires in row i are at most a_i, and at least b_i correspond bijectively to families $(P_1, \cdots, P_r)$ of nonintersecting lattice paths, in which P_i runs from $(-\lambda_i + i, b_i - i)$ to $(-\mu_i + i, a_i - i)$, and P_r shifted by $(-r - d, r)$, does not intersect P_1.

Proposition 4.1. *[6] Suppose S be the shift by $(-a_x, a_y), a_x, a_y \in \mathbb{Z}^+$, a function w on the horizontal edges of $\mathbb{Z}^2$ with the property that for all horizontal edges e and some fixed interminate z, which $w(Se) = z \cdot w(e)$. Let $u = (u_1, \cdots, u_r)$ and $v = (v_1, \cdots, v_r)$ be $r-$tuples of points in $\mathbb{Z}^2$ such that u_i lies to the northwest of u_{i+1} and v_i lies to the northwest of v_{i+1} for $i = 1, 2, \cdots, r-1$. Su_r lies to the northwest of u_1 and Sv_r lies to the northwest of v_1; and u_i lies to the southwest of v_i for $i = 1, 2, \cdots, r$. Then the generating function $\sum_{P=(P_1,\cdots,P_r)} w(P)$, where $P = (P_1, \cdots, P_r)$ of lattice paths in which P_i runs from u_i to v_i and $SP = S(P_1, \cdots, P_r)$ is not intersecting, equals*

$$\sum_{k_1+\cdots+k_r=0} det_{1\leq s,t\leq r}(z^{a_x \frac{k_s^2}{2} - v_{t,x}k_s} GF(\mathcal{P}(u_s \to S^{k_s} v_t); w) \tag{4.14}$$

Theorem 4.2. *[6] Suppose $\lambda = (\lambda_1, \lambda_2, \cdots, \lambda_r)$ and $\mu = (\mu_1, \mu_2, \cdots, \mu_r)$ be $r-$tuples of integers and $d > -r$. The generating function $\sum q^{n(\pi)}$ of the cylindric partition π of shape*

$\lambda/\mu/d$ *which the entries in row* i *are at most* a_i *and at least* b_i *equals*

$$\sum_{k_1+\cdots k_r=0} det_{1\le s,t\le r}(q^{b_s(\lambda_s-s-\mu_t+t-(r+d)k_s)+(\mu_t-t)(s-t+rk_s)+(r+d)(\frac{rk_s^2}{2}+sk_s)}\binom{a_t-b_s+\lambda_s-\mu_t-dk_s}{\lambda_s-\mu_t-s+t-(r+d)k_s}) \tag{4.15}$$

This theorem for the rectangle shape diagram and choose Gaussian kernel would recover the well-known identity[6]

$$\sum_{k_1+\cdots k_r=0} q^{r^2\sum_{i=1}^r \frac{k_i^2}{2}+r\sum_{i=1}^r ik_i} det_{1\le s,t\le r}(q^{t(t-s-rk_s)}\binom{a+c}{c-s+t-rk_s}) = \binom{a+c}{c}_{q^r} \tag{4.16}$$

For the classical case, which $q\to 1, t=q^k\to 1$, fix period $(i,-j)$, a periodic skew Young diagram is determined by two monotone lattice paths from $(0,0)$ to $(i,-j)$, which P_-, P_+(which determine the X-weights and Y-weights simiountanously, in representation theory, it means given fundamental alcove $\lambda_1-\lambda_i\le j, 0\le(\lambda,\alpha_i),(\lambda,\theta^\vee)\le j$, λ is any dominant weight, in general it is in form $\lambda=\xi_1\lambda_1+\cdots\xi_n\lambda_n, \xi_i\ge 0, 1\le i\le n$) A single monotone path has $\binom{i+j}{i}^2$ choices. (The meaning of $\binom{i+j}{i}$ is cleary that it is the Poincare polynomial of the corresponding Grassmannian $Gr(i,i+j)$.) For noncrossing path, which require $P_-\le P_+$, then the Lindstrom-Gessel-Viennot determinant gives the number of admissible pairs. For instance, the width 2 skew Young diagram, the number of noncrossing paths determinant of binomial coefficients

$$\#_{i,j}=det\begin{pmatrix}\binom{i+j}{i} & \binom{i+j}{i+1}\\ \binom{i+j}{i-1} & \binom{i+j}{i}\end{pmatrix} \tag{4.17}$$

thus

$$\#_{i,j}=\binom{i+j}{i}^2-\binom{i+j}{i-1}\binom{i+j}{i+1} \tag{4.18}$$

this is the analogue of Catalan type counting. In order to recover some information of DAHA, the easiest way is do the following replacement,

$$\binom{i+j}{i}\Longrightarrow\binom{i+j}{i}_{q^k} \tag{4.19}$$

where

$$\binom{i+j}{i}_{q^r}=\frac{(1-(q^r)^{i+j})(1-(q^r)^{i+j-1})\cdots(1-(q^r)^{j+1})}{(1-(q^r))(1-(q^r)^2)\cdots(1-(q^r)^i)} \tag{4.20}$$

5 Theta Functions, Metaplectic Multipliers, and Modular Geometry

We now bring together the modular transformation properties of the Dedekind eta function and Jacobi theta functions, focusing on the roots of unity and multiplier systems that emerge from their interplay. We then study how these structures degenerate on even lattices and extend geometrically through logarithmic connections and Siegel modular forms.

We first recall a classical fact concerning the quantization of fermions on a vector bundle. Given a family of Dirac operators parametrized by a moduli space, fermionic quantization does not produce an ordinary function but rather a section of the determinant line bundle associated with the Dirac operator. As one moves around a nontrivial loop in moduli space, this determinant line bundle may acquire a nontrivial holonomy. In the case of elliptic curves, the determinant line bundle is closely related to the Hodge line bundle, and its modular properties are encoded by the Dedekind η-function.

From the physical perspective, the Dedekind η-function arises naturally through zeta-function regularization. Recall the bosonic partition function

$$\mathcal{Z}(q) = q^{-E_0} \prod_{n\geq 1} \frac{1}{1-q^n}, \qquad E_0 = -\frac{1}{24}, \tag{5.1}$$

whose inverse is

$$\eta(\tau) = q^{1/24} \prod_{n\geq 1} (1-q^n). \tag{5.2}$$

it satisfy the differential equation

$$\frac{d}{d\tau} log(\eta(\tau)) = \frac{\pi i}{12} E_2(\tau) \tag{5.3}$$

where

$$E_2(\tau) = 1 - 24 \sum_{n\geq 1} \sigma_1(n) q^n \tag{5.4}$$

equivalently

$$\eta(\tau)' = \frac{\pi i}{12} E_2(\tau)\eta(\tau) \tag{5.5}$$

Under the action of the modular group $SL_2(\mathbb{Z})$, under action T, one has

$$\eta(\tau+1) = e^{\pi i/12}\eta(\tau) \tag{5.6}$$

and under action S, one has

$$\eta\left(-\frac{1}{\tau}\right) = \sqrt{-i\tau}, \eta(\tau). \tag{5.7}$$

The T-transformation therefore introduces a twelfth root of unity, while the S-transformation involves the square root of $-i\tau$, whose multiplier system is ultimately governed by eighth roots of unity. Together with the vacuum-energy contribution $q^{1/24}$, these modular factors imply that the η-function naturally lives in a metaplectic setting characterized by twenty-fourth roots of unity. Geometrically, this reflects the fact that the determinant line bundle possesses a nontrivial modular multiplier rather than a genuine $SL_2(\mathbb{Z})$ action. Thus, the

appearance of the 12-th root of unity originates from the fact that the quantum fermionic partition function whihc is not an ordinary function on moduli space.

In the fermionic path integral, integration over Grassmann variables produces a Pfaffian rather than a determinant. For an antisymmetric operator A,

$$\mathrm{Pf}(A)^2 = \det(A). \tag{5.8}$$

Consequently, fermionic quantization requires a choice of square root of the determinant line bundle. The resulting Pfaffian line bundle carries a finer modular structure, and its holonomies are controlled by the same metaplectic phenomena responsible for the multiplier system of the η-function.

A closely related structure appears in the theory of theta functions and vertex operators. The latter are fundamentally constructed from theta functions, whose modular transformation law takes the form

$$\theta(z,\tau) = \varphi(\tau)\exp\left(-\frac{\pi i c z^2}{c\tau+d}\right)\theta\left(\frac{z}{c\tau+d}, \frac{a\tau+b}{c\tau+d}\right) \tag{5.9}$$

The factor $\varphi(\tau)$ can be expressed through a Gaussian integral and eventually reduces to a quadratic Gauss sum,

$$\varphi(\tau) = \sum_{n \bmod c} e^{-\pi i n^2 d/c} \int_{-\infty}^{\infty} e^{\pi i c^2 y^2(\tau+d/c)}, dy. \tag{5.10}$$

Using

$$\int_{-\infty}^{\infty} e^{\pi i c^2 y^2(\tau+d/c)} dy = \frac{1}{c\left((\tau+d/c)/i\right)^{1/2}}, \tag{5.11}$$

one finds that the modular multiplier is essentially a Gauss sum. Classical number theory shows that such Gauss sums are equal to $c^{1/2}$ multiplied by an eighth root of unity. Hence the same eighth-root phenomenon appearing again, but the meaning is different from η function.

From a geometric viewpoint, theta functions serve as local trivializations of line bundles over abelian varieties and moduli spaces. Their distinguished role is reflected in the heat equation

$$\frac{\partial}{\partial t}\theta(x,it) = \frac{1}{4\pi}\frac{\partial^2}{\partial x^2}\theta(x,it), \tag{5.12}$$

which may be interpreted as the parallel transport of sections under the natural connection on the line bundle. In this sense, theta functions provide the analytic realization of the bundle's trivialization and encode its modular and geometric structure simultaneously.

The appearance of eighth roots of unity ,therefore, not accidental. It has a real physical meaning. In modern physics they manifest themselves as global anomalies, 't Hooft anomalies, invertible TQFT phases, and spin-refined partition functions. Even in lattice models and condensed matter systems, where the gauge group is often discrete rather than continuous, analogous multiplier systems persist through discrete versions of theta functions and quadratic refinements. Thus, the ubiquitous eighth-root structure can be viewed as a universal remnant of "trivilazation" on bundles.

On the DAHA side, for the sharp $q-$ Mellin transfrom defined by the truncated θ function[1]

$$\Psi_k^\sharp = \frac{1}{ig_k}\int_C f(x)\delta_k(x)dx \tag{5.13}$$

in $\Re k > 0$ and $|\Im k| < 2\epsilon$, it could be expressed in

$$\Psi_k^\sharp = (-a\pi)\prod_{j=1}^{\infty}\frac{(1-q^{j+k})(1-q^{j-k-1})}{(1-q^j)^2}\sum_{j=-\infty}^{\infty} q^{(k-j)^2/4} \tag{5.14}$$

Moreover, it could be expressed in a more algebraiclly way[1]

$$\sum_{j=0}^{\infty} q^{(k-j)^2/4}\frac{1-q^{j+k}}{1-q^k}\prod_{l=1}^{j}\frac{1-q^{l+2k-1}}{1-q^l} = \prod_{j=1}^{\infty}\frac{1-q^{j+k}}{1-q^j}(\sum_{j=-\infty}^{\infty} q^{(k-j)^2/4}) \tag{5.15}$$

The left-hand side is a classical Jackson sum. When $k = \frac{1}{2}$, it contains the prefactor $2q^{1/16}$. The appearance of the fractional exponent $(1/16)$ suggests considering modular objects with multipliers determined by roots of unity. To treat the cases corresponding to 16-th, 12-th, and 8-th roots of unity in a uniform manner, it is natural to work in the setting of multipliers of order dividing 48, since

$$48 = \mathrm{lcm}(16, 12, 8). \tag{5.16}$$

This motivates the search for modular forms carrying a general (48)-th root-of-unity multiplier system.

Since, the metaplectic group $(Mp_2(F))$ is realized as a group of unitary operators on $L^2(F)$, fitting into the central extension

$$1 \longrightarrow T \longrightarrow Mp_2(F) \longrightarrow SL_2(F) \longrightarrow 1. \tag{5.17}$$

It is well known that this extension represents a nontrivial element of order two in the cohomology group

$$H^2\left(SL_2(F), T\right). \tag{5.18}$$

Our goal is to construct modular forms whose multiplier systems take values in a genuinely non-split central extension. More precisely, we consider the metaplectic extension

$$1 \longrightarrow \mu_2 \longrightarrow Mp_2(\mathbb{Z}) \longrightarrow SL_2(\mathbb{Z}) \longrightarrow 1, \tag{5.19}$$

and tensor its canonical multiplier with an independent character of order (24), arising for instance from the Dedekind η-multiplier. The resulting multiplier system takes values in a central extension whose cocycle class combines the metaplectic 2-torsion with the η-character and therefore does not descend to a character of $SL_2(\mathbb{Z})$. Consequently, the associated modular forms are naturally defined on the metaplectic cover rather than on $SL_2(\mathbb{Z})$ itself. In particular, combining the metaplectic multiplier of order (2) with the Dedekind η-multiplier of order (24) yields a multiplier system taking values in roots of unity of order dividing

$$\mathrm{lcm}(2, 24) = 48. \tag{5.20}$$

Thus the natural setting for these modular forms is a 48-fold multiplier system on $Mp_2(\mathbb{Z})$, providing a uniform framework for the 8-th, 12-th, 16-th, and 24-th root phenomena that arise in the corresponding q-series.

Recall some properties of η function maybe used later

$$\frac{\eta(\tau)}{\eta(\tau/2)} = q^{1/48} \sum_{n\geq 0} \frac{q^{n/2}}{(q)_n} \tag{5.21}$$

$$\frac{\eta(\tau/2)\eta(2\tau)}{\eta^2(\tau)} = q^{1/48} \sum_{n\geq 0} (-1)^n \frac{q^{n/2}}{(q)_n} \tag{5.22}$$

Before proceeding, we introduce a unified notation that will be used throughout this paper. The Jacobi theta function

$$\theta \begin{bmatrix} a \\ b \end{bmatrix} (z,\tau) = \sum_{n\in\mathbb{Z}} \exp\Big(\pi i(n+a)^2\tau + 2\pi i(n+a)(z+b)\Big) \tag{5.23}$$

explicitly,

$$\theta_3(z,\tau) = \theta \begin{bmatrix} 0 \\ 0 \end{bmatrix} (z,\tau) \tag{5.24}$$

$$\theta_4(z,\tau) = \theta \begin{bmatrix} 0 \\ \frac{1}{2} \end{bmatrix} (z,\tau) \tag{5.25}$$

$$\theta_2(z,\tau) = \theta \begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix} (z,\tau) \tag{5.26}$$

$$\theta_1(z,\tau) = \theta \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} (z,\tau) \tag{5.27}$$

define

$$\theta \begin{bmatrix} \epsilon \\ \delta \end{bmatrix} := \theta \begin{bmatrix} \epsilon/2 \\ \delta/2 \end{bmatrix}, \qquad \epsilon, \delta \in \{0,1\} \tag{5.28}$$

and define the parity

$$(-1)^{\epsilon\delta} \tag{5.29}$$

then

$$\vartheta_3 = \theta \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \text{even} \qquad \vartheta_4 = \theta \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \text{even} \qquad \vartheta_2 = \theta \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \text{even} \qquad \vartheta_1 = \theta \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{odd} \tag{5.30}$$

thus $\theta_2, \theta_3, \theta_4$ are corresponding to three even structures, and θ_1 attachs to odd spin structure.

$$\vartheta_3(\tau) = \sum_{n\in\mathbb{Z}} q^{n^2/2} \tag{5.31}$$

$$\vartheta_4(\tau) = \sum_{n\in\mathbb{Z}} (-1)^n q^{n^2/2} \tag{5.32}$$

$$\vartheta_2(\tau) = \sum_{n\in\mathbb{Z}} q^{(n+\frac{1}{2})^2/2} \tag{5.33}$$

$$\vartheta_1(\tau) = 0 \tag{5.34}$$

under S, T transformations

$$S : \tau \mapsto -\frac{1}{\tau} \quad T : \tau \mapsto \tau + 1 \quad S^2 = (ST)^3 = -I \tag{5.35}$$

they transform as

$$S : \theta_2 \leftrightarrow \theta_4, \quad \theta_3 \mapsto \theta_3 \tag{5.36}$$

$$T : \theta_2 \mapsto e^{\pi i/4}\theta_2, \quad \theta_3 \leftrightarrow \theta_4 \tag{5.37}$$

First, we construct

$$\mathcal{G}(\tau) = \eta(\tau)\theta_{10}(2\tau) = \eta(\tau)\vartheta_2(2\tau) \tag{5.38}$$

under this definition, the shifted theta function contributes ζ_8 but with a nontrivial 2-cocyle under rescaling. since under scale $\tau \mapsto 2\tau$, the Weil representation twists so that, the T-action becomes $e^{\pi i/2}$ of order 4, but the interacts with η nontrivially. thus

$$lcm(24, 8) \times 2 = 48 \tag{5.39}$$

it lives in the category

$$\mathrm{Rep}\left(Mp_2(\mathbb{Z})\right), \tag{5.40}$$

and tensor its objects with the Weil representation ρ_{A_1}. Rearrange it by

$$\theta_{10}(\tau) = 2\frac{\eta(2\tau)^2}{\eta(\tau)} \tag{5.41}$$

then

$$\mathcal{G}(\tau) = 2\frac{\eta(\tau)\eta(4\tau)^2}{\eta(2\tau)} \tag{5.42}$$

it is an arithmetic ζ_{48} which created from combining η-multiplier with metaplectic square root.

Second, we define

$$\mathcal{F}(\tau) = \eta(\tau)\frac{\theta(\tau + \frac{1}{2})}{\theta(\tau)} \tag{5.43}$$

It arises as the difference of two theta characteristics and can be interpreted as a logarithmic derivative of a theta line bundle. The associated multiplier system lives in a projective Weil

representation. Geometrically, this yields a ζ_{48}-valued structure, originating from a choice of nontrivial trivialization of the underlying theta line bundles.

We would rewrite $\mathcal{G}$ using Hurwitz quaternions and Hurwitz order. Recall

$$\mathcal{H} = \mathbb{Z}[1, i, j, \frac{1+i+j+k}{2}] \tag{5.44}$$

elements in from

$$q = a + bi + cj + dk \tag{5.45}$$

the norm is

$$N(q) = q\bar{q} = a^2 + b^2 + c^2 + d^2 \tag{5.46}$$

since the half-integer point $\frac{1+i+j+k}{2}$, the lattice is not $\mathbb{Z}^4$, but rather D_4^*(the dual/weight lattice of D_4 ($D_4 = \{x = (x_1, x_2, x_3, x_4) \in \mathbb{Z}^4 | x_1 + x_2 + x_3 + x_4 = 0 \, (mod \, 2)\}$)), The integral norm vectors from the root lattice D_4. Under the theta function defined under quadratic forms, it has form

$$\Theta_H(\tau) = \sum_{q \in \mathcal{H}} q^{N(q)} \tag{5.47}$$

and it have relation

$$\Theta_H(\tau) = \theta_3(\tau)^4 = 1 + 8q + 24q^2 + 32q^3 + 24q^4 + \cdots \tag{5.48}$$

since the D_4 algebra has famous triality

$$8_v, \quad 8_s, \quad 8_c \tag{5.49}$$

thus, the theta representation therefore splits into several pieces. since the discriminant group is

$$D_4^*/D_4 \cong (\mathbb{Z}/2\mathbb{Z})^2 \tag{5.50}$$

thus have four coset

$$0, v, s, c \leftrightarrow \Theta_0, \Theta_v, \Theta_s, \Theta_c \tag{5.51}$$

$$D_4^* = D_4 \cup (D_4 + \omega_v) \cup (D_4 + \omega_s) \cup (D_4 + \omega_c) \tag{5.52}$$

they are not independent, but transform among themselves under $SL_2(\mathbb{Z})$. their relation with jocobian theta functions are

$$\Theta_0 = \frac{1}{2}(\theta_3^4 + \theta_4^4) \tag{5.53}$$

$$\Theta_v = \frac{1}{2}(\theta_3^4 - \theta_4^4) \tag{5.54}$$

$$\Theta_s = \frac{1}{2}\theta_2^4 \tag{5.55}$$

$$\Theta_c = \frac{1}{2}\theta_2^4 \tag{5.56}$$

the $\theta_2^4, \theta_3^4, \theta_4^4$ are likewise permuted by modular transformations, the Jacobi's identity

$$\theta_3^4 = \theta_2^4 + \theta_4^4 \tag{5.57}$$

is essentially the character identity

$$8_v = 8_s + 8_c \tag{5.58}$$

Particular, for A_1 case/Hurwitz order,

$$Q(n) = n^2 \tag{5.59}$$

then

$$\theta_{A_1}(\tau) = \sum_n q^{n^2} = \theta_3(2\tau) \tag{5.60}$$

and

$$\theta_{A_1}(\tau + \frac{1}{2}) = \sum_n (-1)^{n^2} q^{n^2} = \theta_4(2\tau) \tag{5.61}$$

then

$$\mathcal{F}(\tau) = \eta(\tau)\frac{\theta_4(2\tau)}{\theta_3(2\tau)} \tag{5.62}$$

using the jocobian identity of theta functions, and define

$$k = \frac{\theta_2^2}{\theta_3^2} \tag{5.63}$$

then

$$1 = (\frac{\theta_4}{\theta_3})^4 + (\frac{\theta_2}{\theta_3})^4 = (1 - k^2) + k^2 \tag{5.64}$$

thus define

$$k' = \sqrt{1 - k^2} \tag{5.65}$$

then

$$\frac{\theta_4}{\theta_3} = (1 - k^2)^{1/4} = \sqrt{k'} \tag{5.66}$$

it is well related with Hurwitz elliptic curve.

5.1 Degenerate case: in even lattice

For even lattices such as $A_1^{\oplus n}$, (D_4), and E_8, the function $\mathcal{F}(\tau)$ degenerates under explicit calculation to the Dedekind eta function:

$$\mathcal{F}(\tau) = \eta(\tau). \tag{5.67}$$

This degeneration reflects cancellations in the theta-characteristic decomposition: shifted theta functions are related by nontrivial multiplier phases rather than independent contributions, so that their ratios become trivial in the resulting modular product, leaving only the η-factor.

On an even lattice, consider the set

$$\mathcal{S} = \left\{\eta(\tau),\, \vartheta_2(2\tau),\, \eta(\tau)\vartheta_2(2\tau),\, \eta(\tau)^2\vartheta_2(2\tau)\right\}. \tag{5.68}$$

Lemma 5.1. *The elements in $\mathcal{S}$ generate a graded subalgebra of the ring of Jacobi/metaplectic modular forms*

$$\mathcal{A} = \bigoplus_{k\in\frac{1}{2}\mathbb{Z}} \mathcal{M}_k(\Gamma_0(4), \rho), \tag{5.69}$$

where ρ is the Weil representation associated with the quadratic form modulo 2. The grading is compatible with multiplication:

$$\mathcal{M}_k \cdot \mathcal{M}_l \subset \mathcal{M}_{k+l}. \tag{5.70}$$

These functions arise from the Weil representation of the metaplectic group $Mp_2(\mathbb{Z})$ and correspond to theta–eta combinations attached to the underlying even lattice structure.

$$\vartheta_2(\tau, z) = 2q^{1/8}y^{1/2}\prod_{n\geq 1}(1-q^n)(1-yq^n)(1-y^{-1}q^n) \quad y = e^{2\pi iz} \tag{5.71}$$

at $z = 1/2$, $y = -1$, then

$$\vartheta_2(\tau, \frac{1}{2}) = 2q^{1/8}\prod_{n\geq 1}(1-q^n)(1+q^n)^2 = 2\frac{\eta(2\tau)^2}{\eta(\tau)} \tag{5.72}$$

if is the Remond Mojorana fermionic sector. (M-NS^+:$\sqrt{\vartheta_3/\eta}$, M-NS^-:$\sqrt{\vartheta_4/\eta}$, M-R:$\sqrt{\vartheta_2/2\eta}$)

start with A_1 lattice,

$$L = A_1 = \sqrt{2}\mathbb{Z} \tag{5.73}$$

the bilinear form on it is

$$(\alpha, \alpha) = 2 \tag{5.74}$$

then the dual lattice is

$$L^* = \frac{\alpha}{2}\mathbb{Z} \tag{5.75}$$

then

$$L^*/L \cong \mathbb{Z}_2 \tag{5.76}$$

here, we have two coset

$$\bar{0} : 0 + L \quad \bar{1} : \frac{\alpha}{2} + L \tag{5.77}$$

these two coset correspondence with

$$\Theta_{\bar{0}}(\tau) = \sum_{n\in\mathbb{Z}} q^{n^2} = \vartheta_3(2\tau) \tag{5.78}$$

$$\Theta_{\bar{1}}(\tau) = \sum_{n\in\mathbb{Z}} q^{(n+\frac{1}{2})^2} = \vartheta_2(2\tau) \tag{5.79}$$

For lattice VOA of A_1, it has two irreducible modules

$$V_{\bar{0}} \quad V_{\bar{1}} \tag{5.80}$$

their characters are

$$\chi_{\bar{0}} = \frac{\Theta_{\bar{0}}(\tau)}{\eta(\tau)} \quad \chi_{\bar{1}} = \frac{\Theta_{\bar{1}}(\tau)}{\eta(\tau)} \tag{5.81}$$

thus ϑ_2 is character of the odd coset modules.

Recall that the Dedekind eta function $\eta(\tau)$ appears in quantum field theory as the renormalized vacuum-energy contribution of the bosonic oscillator modes. In contrast, the theta function $\vartheta_2(2\tau)$ encodes the shifted lattice momentum sector of the A_1 lattice theory. The collection

$$\mathcal{S} = \left\{\eta(\tau),\, \vartheta_2(2\tau),\, \eta(\tau)\vartheta_2(2\tau),\, \eta(\tau)^2\vartheta_2(2\tau)\right\}$$

forms a four-dimensional algebra closed under the $\mathbb{Z}_2 \times \mathbb{Z}_2$ fusion rules associated with the level-one A_1 lattice vertex operator algebra.

The corresponding lattice VOA is

$$V_{A_1} = M(1) \otimes \mathbb{C}[A_1], \tag{5.82}$$

where

$$M(1) = \mathbb{C}[\alpha_{-1}, \alpha_{-2}, \ldots] \tag{5.83}$$

is the rank-one Heisenberg Fock space. $\mathbb{C}[A_1]$ is the group algebra of the root lattice $A_1 = \sqrt{2}, \mathbb{Z}$. The two factors correspond physically to oscillator excitations and lattice momentum states, respectively.

Its graded character is

$$\mathrm{ch}_{M(1)} = \prod_{n\geq 1} \frac{1}{1-q^n} = \frac{q^{-1/24}}{\eta(\tau)} \tag{5.84}$$

showing that the Dedekind eta function is the universal contribution of the bosonic oscillator vacuum. In quantum field theory this factor arises from zeta-function regularization of the infinite collection of harmonic oscillators and may therefore be regarded as the renormalized bosonic vacuum-energy contribution.

The lattice sector contributes through the theta series $\Theta_{\bar{0}}(\tau)$ together with its shifted coset $\Theta_{\bar{1}}(\tau)$. The function $\vartheta_2(2\tau)$ therefore records states carrying half-integral lattice momentum and corresponds to the nontrivial coset of the discriminant group

$$A_1^\vee / A_1 \cong \mathbb{Z}_2. \tag{5.85}$$

Consequently, $\eta(\tau)$ and $\vartheta_2(2\tau)$ encode two fundamental ingredients of the theory:

$$\eta(\tau) \longleftrightarrow \text{oscillator vacuum sector} \quad \vartheta_2(2\tau) \longleftrightarrow \text{shifted lattice momentum sector} \tag{5.86}$$

Taking products of these objects produces mixed sectors combining oscillator and momentum degrees of freedom. In particular, $\eta(\tau)\vartheta_2(2\tau)$ represents a simultaneous contribution from both structures, while $\eta(\tau)^2\vartheta_2(2\tau)$ may be viewed as a higher-weight modular descendant generated by repeated multiplication with the bosonic vacuum factor.

From the perspective of modular representation theory, the theta functions form basis vectors of the Weil representation attached to the discriminant form

$$(A_1^\vee/A_1, q) \cong (\mathbb{Z}_2, x^2/2). \tag{5.87}$$

The modular transformations S and T act through the metaplectic representation of $SL_2(\mathbb{Z})$, and the theta constants furnish a finite-dimensional module for this action. The eta function supplies the one-dimensional determinant multiplier system, so that products of η and theta functions naturally generate modular objects transforming under the same metaplectic symmetry.

Therefore the collection

$$\mathcal{S} = \{\eta(\tau), \vartheta_2(2\tau), \eta(\tau)\vartheta_2(2\tau), \eta(\tau)^2\vartheta_2(2\tau)\} \tag{5.88}$$

may be regarded as a finite-dimensional shadow of the full A_1 lattice VOA. It captures the interaction between the bosonic oscillator sector and the nontrivial lattice coset sector, reflects the $\mathbb{Z}_2$ structure of the discriminant group, and realizes a small graded commutative algebra inside the ring of modular forms and theta constants. In physical language, this algebra describes elementary partition-function building blocks of a spin-structure-resolved two-dimensional conformal field theory. In algebraic language, it is a subalgebra generated by theta constants in the Weil representation of the A_1 discriminant form. In vertex-algebraic language, it is a finite modular image of the fusion structure underlying the level-one A_1 lattice vertex operator algebra.

5.2 Connections and Siegel modular forms

Now we come back to consider the set

$$\{\mathcal{F}, \mathcal{G}\} \tag{5.89}$$

first, $\mathcal{F}$ sits in a Jacobi-theta three-vector, we set

$$\tilde{\Theta}(\tau) = \begin{pmatrix} \theta_2(\tau) \\ \theta_3(\tau) \\ \theta_4(\tau) \end{pmatrix} \tag{5.90}$$

the S, T transformations under this setting are

$$\rho(T) = \begin{pmatrix} e^{\pi i/4} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \rho(S) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \tag{5.91}$$

combine the differential equation satisfied by θ and η function, we get

$$\frac{\mathcal{F}'}{\mathcal{F}} = \frac{\pi i}{12} E_2 - \frac{\pi i}{4}\theta_2^4 \tag{5.92}$$

Function $\mathcal{F}$'s logarithmic connection is

$$\nabla_{\mathcal{F}} = d - \frac{d\mathcal{F}}{\mathcal{F}} = d - (\frac{\pi i}{12} E_2 - \frac{\pi i}{4}\theta_2^4) d\tau \tag{5.93}$$

We use the previous nondegenerate parameterization $\theta_2^4 = \lambda\theta_3^4$, we would get

$$\theta_3^4 = \frac{16}{\pi i} \frac{\lambda'}{\lambda(1-\lambda)} \tag{5.94}$$

thus, the connection of $\mathcal{F}$ becomes

$$\nabla_{\mathcal{F}} = d - [\frac{\pi i}{12} E_2 - 4\frac{\lambda'}{\lambda(1-\lambda)}]d\tau \tag{5.95}$$

clear, from this expression, there are three singularities $\lambda = 0, 1, \infty$.

Now look the relation between function $\mathcal{G}$ and Siegel modular forms. function $\mathcal{G}$ satisfy

$$\frac{\mathcal{G}'}{\mathcal{G}} = \frac{\pi i}{12}(E_2(\tau) + 4E_2(2\tau) - 4E_2(4\tau)) \tag{5.96}$$

Function $\mathcal{F}$'s logarithmic connection is

$$\nabla_{\mathcal{G}} = d - \frac{\pi i}{12}[E_2(\tau) + 4E_2(2\tau) - 4E_2(4\tau)]d\tau \tag{5.97}$$

take a closer look on $\mathcal{F}$, take

$$x(\tau) = \frac{\vartheta_4(2\tau)}{\vartheta_3(2\tau)} \tag{5.98}$$

it is a modular function on level 4 $\Gamma_0(4)$.By previous parameterization

$$x(\tau) = \frac{1-k}{1+k}, \quad k = \frac{\vartheta_2(\tau)^2}{\vartheta_3(\tau)^2} \tag{5.99}$$

when τ is a CM point, which

$$a\tau^2 + b\tau + c = 0 \quad a, b, c \in \mathbb{Z} \tag{5.100}$$

then

$$t \in K = \mathbb{Q}(\sqrt{-N}) \tag{5.101}$$

for some positive integer N. It's primitive skew-Hermitian form

$${}^tMB = BM^* \quad B^* = -B \tag{5.102}$$

and related quadternion algebra

$$D = (\frac{-1, -N}{\mathbb{Q}}) \tag{5.103}$$

are

$$\langle K, B, D\rangle = \langle \mathbb{Q}(-\sqrt{N}), \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, (\frac{-1, -N}{\mathbb{Q}})\rangle \tag{5.104}$$

The Fricke involution on $\Gamma_0(4)$ is

$$\mathcal{W}_4 : \tau \mapsto -\frac{1}{4\tau} \tag{5.105}$$

which translated ϑ_3, ϑ_4 into each other. Actually, pair $(\vartheta_3, \vartheta_4)$ forms a two dimensional spinor representation of the metaplectic double cover $Mp_2(\mathbb{Z}) \to SL_2(\mathbb{Z})$. The associated quaternion algebra D be identified with the even Clifford algebra $Cl^0(B) \cong Cl^0(1,1) \cong D$, where B is the underlying skew-Hermitian form. This additional Cliford module appeared previous section where we built $\mathfrak{osp}(1|2)$ by extension of the A_1.

$\mathcal{G}(\tau)$ is a genus one object, while Siegel modular forms live on

$$\mathfrak{H}_g = \{\Omega = \Omega^T, \Im\Omega > 0\} \tag{5.106}$$

so we give one possible geometric explaination of $\mathcal{G}(\tau)$. The genus g theta function is

$$\theta \begin{bmatrix} \epsilon \\ \delta \end{bmatrix} (\Omega) = \sum_{n \in \mathbb{Z}^\vartheta} e^{\pi i (n+\epsilon)^T \Omega (n+\epsilon) + 2\pi i (n+\epsilon)^T \delta} \tag{5.107}$$

in this form

$$\mathcal{G}(\tau) = \eta(\tau) \frac{\theta \begin{bmatrix} 0 \\ 1/2 \end{bmatrix} (\Omega)}{\theta \begin{bmatrix} 0 \\ 0 \end{bmatrix} (\Omega)} \tag{5.108}$$

Luckily, for genus 1, $\eta^{24} = \triangle$, is related with the determinant of the Hodge bundle. For higher genus(here means the Ω which determined by the genus of modular space), there should have a factor of determinant of the Hodge bundle(Igusa modular froms). Thus, considering

$$\mathcal{A}_g = Sp(2g, \mathbb{Z}) \setminus \mathfrak{H}_g \tag{5.109}$$

then each point corresponds to a principally polarized abelian variety

$$A_\Omega = \mathbb{C}^g / (\mathbb{Z}^g + \Omega \mathbb{Z}^g) \tag{5.110}$$

thus

$$\mathcal{G}_g \in H^0(\mathcal{A}_g, \lambda^{1/2} \otimes \mathcal{O}(\Theta_\alpha - \Theta_\beta)) \tag{5.111}$$

where Θ is the theta divisor, satisfied $\Theta = \{e \in \mathcal{A}_\Omega | \theta(z, \Omega) = 0\}$, for our genus one case, the divisor is just a single point

$$\Theta = \{\frac{1+\tau}{2}\} \tag{5.112}$$

up to lattice translation. For the function we constructed

$$\eta \in H^0(\mathcal{A}_1, \lambda^{1/2}) \tag{5.113}$$

$$\mathcal{F} \in H^0(\mathcal{A}_1, \lambda^{1/2} \otimes \mathcal{O}(\Theta_2)) \tag{5.114}$$

$$\mathcal{G} \in H^0(\mathcal{A}_1, \lambda^{1/2} \otimes \mathcal{O}(\Theta_4 - \Theta_3)) \tag{5.115}$$

$$\mathcal{F}\mathcal{G} \in H^0(\mathcal{A}_1, \lambda^{1/2} \otimes \mathcal{O}(\Theta_2 + \Theta_4 - \Theta_3)) \tag{5.116}$$

$$\tag{5.117}$$

(I wanna address here that here the discriminant group $A_2^*/A_2 \cong \mathbb{Z}_3$, hence there are naturally three theta functions, have quadternion algebra on it.) and they correspond to translating the divisor by the four 2-torsion points of E_τ

$$E_\tau[2] = \{0, \frac{1}{2}, \frac{\tau}{2}, \frac{1+\tau}{2}\} \cong (\mathbb{Z}/2\mathbb{Z})^2 \tag{5.118}$$

P.S. 1. At this point, we would like to address a possible source of confusion. It is well known that the DAHA is intrinsically associated with a double torus structure, reflected algebraically by its two deformation parameters and geometrically by two elliptic directions. One may therefore wonder why our six-dimensional construction appears to involve an additional elliptic degree of freedom. The answer lies in the geometric realization rather than in the algebra itself. Our construction of the set $\mathcal{S} = \{\eta, \mathcal{F}, \mathcal{G}, \mathcal{FG}\}$ already exhibits three independent theta-function structures. While the first two elliptic directions correspond to the intrinsic double-affine geometry of DAHA, the third arises naturally when these modular objects are interpreted as sections of vector bundles over elliptic curves or, more generally, over principally polarized abelian varieties. In other words, the additional elliptic structure is not an extra deformation parameter of DAHA itself, but is introduced by the geometry required to realize its representations. From this perspective, the six-dimensional picture is not merely the product of two elliptic curves. Rather, the modular realization naturally incorporates a third elliptic component associated with the Hodge bundle, the theta line bundle, or equivalently the spin structure carried by the corresponding theta functions. Consequently, although DAHA is fundamentally a double-affine object, its geometric realization in terms of vector bundles and theta functions naturally involves three elliptic geometries.

6 From Theta Kernels to DAHA Knots

We explore possible connections between modular theta kernels and DAHA knot invariants, using the torus knot $T(3,2)$ as example. Our aim is to identify common structures linking q-series, Fourier kernels, and representation theory, while distinguishing explicit identities from conjectural geometric and physical interpretations.

The Knots appearing in DAHA are all elliptic knots. We take one here as an example. Recall $T(3,2)$ has braid group description

$$T(3,2) = \langle x, y | x^3 = y^2 \rangle \tag{6.1}$$

for x^3 whose monodromy has order 6. We introduce

$$\omega_0 = e^{2\pi i/3} \leftrightarrow \mathbb{Q}(\sqrt{-3}) \tag{6.2}$$

therefore, the Eisenstain lattice is

$$\omega = \frac{-1+\sqrt{-3}}{2} \tag{6.3}$$

thus, it's ring of integers $\mathbb{Z}(\omega)$ is the A_2 lattice. now we derive the Torus knot $T(3,2)$ from the function $\mathcal{F}$

$$\mathcal{F}(\tau) = \eta(\tau)\frac{\vartheta_4(\tau)}{\vartheta_3(\tau)} = q^{1/24}\prod_{n=1}^{\infty}(1-q^n)(\frac{1-q^{2n-1}}{1+q^{2n-1}})^2 \tag{6.4}$$

then, the natural refinement is

$$\mathcal{F}(\tau, x) = \eta(\tau)\prod_{n\geq 1}\frac{(1-xq^{2n-1})(1-x^{-1}q^{2n-1})}{(1+xq^{2n-1})(1+x^{-1}q^{2n-1})} \tag{6.5}$$

570 if expand $\mathcal{F}$ in forms of $1, x+x^{-1}, x^2+x^{-2}, \cdots$, under impose $x^3=1, x^5=1, x^7=1$

$$\mathcal{F}(\tau,x)_3 = (1-4q^{1/2}+7q+\cdots)+(-4q^{1/2}+8q+\cdots)(x+x^{-1}) \tag{6.6}$$

$$\mathcal{F}(\tau,x)_5 = (1-4q^{1/2}+7q+\cdots)+(-4q^{1/2}+8q+\cdots)(x+x^{-1})+(2q+\cdots)(x^2+x^{-2}) \tag{6.7}$$

$$\mathcal{F}(\tau,x)_7 = (1-4q^{1/2}+7q+\cdots)+(-4q^{1/2}+8q+\cdots)(x+x^{-1})+(2q+\cdots)(x^2+x^{-2})+(-2q^{3/2+\cdots})(x^3+ \tag{6.8}$$

571 take log of orginal $\mathcal{F}$

$$log\,\mathcal{F} = \frac{1}{24} log\, q + \sum_{n\geq 1}\log(1-q^n) + 2\sum_{n\geq 1}\log\frac{1-q^{2n-1}}{1+q^{2n-1}} \tag{6.9}$$

572 set $z=q^{2n-1}$, then

$$\log\mathcal{F} = -\sum_{n,m}\frac{q^{mn}}{m} - 4\sum_{n,r}\frac{q^{(2r+1)(2n-1)}}{2r+1} \tag{6.10}$$

$$= \sum_{s\geq 1} a(s) q^s \tag{6.11}$$

573 where

$$a(s) = -\sum_{d|s}\frac{1}{d} - 4\sum_{d|s, d\,odd, s/d\,odd}\frac{1}{d} \tag{6.12}$$

574 suppose Hilbert space carries an integer charge $Q\in\mathbb{Z}$, thus instand tracing of L_0, we take
575 trace including the charge Q,

$$q^{L_0}x^Q \tag{6.13}$$

576 define

$$Z(q,x) = \sum_Q Z_Q(q)x^Q \tag{6.14}$$

577 suppose $x^N=1$, take $\omega = e^{2\pi i/N}$, then

$$\frac{1}{N}\sum_{j=0}^{N-1}\omega^{-jk}Z(\omega_j) = \sum_Q Z_Q(\frac{1}{N}\sum_j \omega^{j(Q-k)}) \tag{6.15}$$

578 use

$$\frac{1}{N}\sum_j \omega^{j(Q-k)} = \delta_{Q,k(mod\,N)} \tag{6.16}$$

579 thus

$$A_k(q) = \frac{1}{N}\sum_{j=0}^{N-1}\omega^{-jk}Z(q,\omega^j) \tag{6.17}$$

580 for a compact boson

$$J(z) = i\partial\phi(z) \quad Q = \oint J \tag{6.18}$$

the canonical quantization gives

$$L_0 = \frac{1}{2}Q^2 + \sum_{n>0} a_{-n}a_n \tag{6.19}$$

if set

$$Z(q,x) = \sum_{m\in\mathbb{Z}} q^{Q^2/2}x^m \tag{6.20}$$

set $x = \omega^j$, then

$$Z(q,\omega^j) = \sum_{m\in\mathbb{Z}} q^{Q^2/2}e^{2\pi ijm/N} \tag{6.21}$$

this exactly the definition of theta function with characteristic

$$\vartheta\begin{bmatrix} j/N \\ 0 \end{bmatrix}(0,\tau) \tag{6.22}$$

under this, the function $\mathcal{F}$ have nature extension as

$$\mathcal{F}(\tau) = \eta(\tau)\frac{\vartheta\begin{bmatrix} j/N \\ 0 \end{bmatrix}(0,\tau)}{\vartheta_3(\tau)} \tag{6.23}$$

For instance, under $x^3 = 1$

$$x + x^{-1} = -1 \tag{6.24}$$

thus

$$x^m + x^{-m} = \begin{cases} 2 & 3 \mid m \\ -1 & 3 \nmid m \end{cases} \tag{6.25}$$

thus

$$\log \mathcal{F} = \log \eta + 2\sum_{m\ odd\ 3\nmid m} \frac{q^m}{m(1-q^{2m})} - 4\sum_{m\ odd\ 3\mid m} \frac{q^m}{m(1-q^{2m})} \tag{6.26}$$

Recall the logarithm of the colored Jones/HOMFLY partition function has the universal form

$$\log Z = \sum_{m\geq 1} = \frac{1}{m}\frac{q^m}{1-q^m}\frac{(x^{pm}-x^{-pm})(x^{qm}-x^{-qm})}{x^m - x^{-m}} \tag{6.27}$$

compare this formula with the $\log \mathcal{F}$, the difference is

$$\frac{2}{1+q^m} \quad V.S. \quad E(m) = \frac{(x^{pm}-x^{-pm})(x^{qm}-x^{-qm})}{x^m - x^{-m}} \tag{6.28}$$

since

$$\frac{2}{1+q^m} = 2\frac{1-q^m}{1-q^{2m}} \tag{6.29}$$

Recall the super trace of fermion

$$\{\psi, \psi^\dagger\} = 1 \tag{6.30}$$

with Fock space

$$|0\rangle \quad \psi^\dagger|0\rangle \tag{6.31}$$

$$Str(q^H) = Tr((-1)^F q^H) = 1 - q^m, \quad H = N(N = 0, 1) \tag{6.32}$$

for bosonic oscillator,

$$[a_m, a_{-m}] = 1 \quad H = N(N = 0, 1, 2 \cdots) \tag{6.33}$$

thus

$$Z_m = Tr\, q^H = \frac{1}{1 - q^m} \tag{6.34}$$

thus this term means that

$$\frac{1}{1 - q^{2m}} \tag{6.35}$$

is the bosonic partition function under even modes.

$$1 - q^m \tag{6.36}$$

is the super of fermion partition. The extra factor 2 is the degeneracy of the vacuum, both have fermion and boson. This whole factor more nature appear in $\mathfrak{osp}(1|2)$. Since $2/(1 + q^m)$ likes the partition, let's try to rewrite it, set $q = e^{2u}$, then

$$\frac{2}{1 + q^m} = 1 - \tanh(mu) = e^{-mu}\mathrm{sech}(mu) \tag{6.37}$$

recall the fourier transformation

$$\mathscr{F}(\mathrm{sech}(ax)) = \frac{\pi}{a}\mathrm{sech}(\frac{\pi k}{2a}) \tag{6.38}$$

thus

$$\mathscr{F}(e^{-mu}\mathrm{sech}(mu)) = \frac{\pi}{u}\mathrm{sech}(\frac{\pi(k + i)}{2u}) \tag{6.39}$$

it means that this kernel is self-dual under Fourier transformation. Also

$$\frac{d}{du}\tanh(mu) = m\,[sech]^2(mu) \tag{6.40}$$

is the characteristic shape of a Fermi surface.

now we from

$$E(m) = \frac{(x^{pm} - x^{-pm})(x^{qm} - x^{-qm})}{x^m - x^{-m}} \tag{6.41}$$

take $x = e^u$ rewrite it

$$E(m) = \frac{e^{(p+q)mu} - e^{-(p+q)mu}}{e^{mu} - e^{-mu}} - \frac{e^{(p-q)mu} - e^{-(p-q)mu}}{e^{mu} - e^{-mu}} \tag{6.42}$$

for $q = e^u, q^m = e^{mu}$, it becomes the difference of quantum integers

$$E(m) = [m(p + q)]_q - [m(p - q)]_q \tag{6.43}$$

rewrite it

$$E(m) = 2\cosh(mpu)\frac{\sinh(mpu)}{\sinh(u)} \tag{6.44}$$

clear this knots related thing have nothing to do with the thing we made before, but if we mutiply the fractor $2/(1+q^m)$ to it

$$\frac{2}{1+q^m}E(m) = e^{-mu}\frac{cosh(mpu)}{cosh(mu)}\frac{2sinh(mpu)}{sinh(u)} \tag{6.45}$$

the fourier transform of cosh/cosh term is

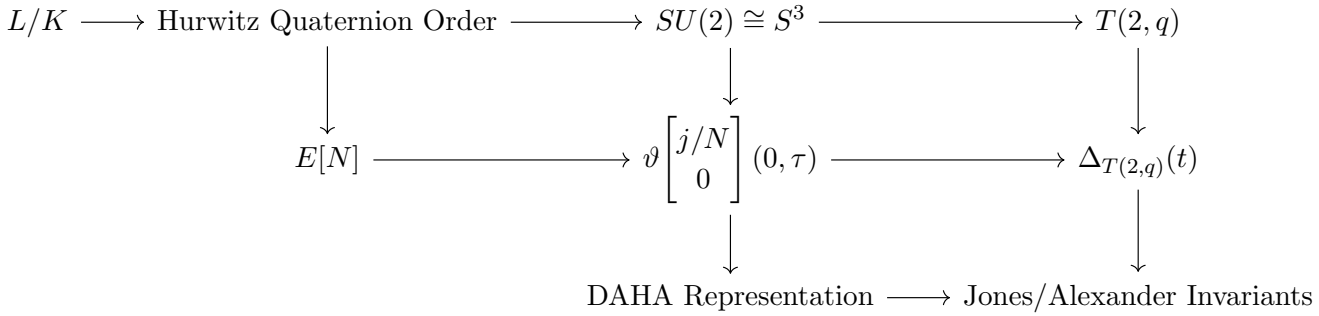
$$\mathscr{F}(\frac{cosh(mpu)}{cosh(mu)}) = \frac{1}{2}(\mathscr{F}\frac{e^{pu}}{coshu} + \mathscr{F}\frac{e^{-pu}}{coshu}) = \frac{1}{2}(sech(k-ip) + sech(k+ip)) \tag{6.46}$$

actually this frac is the simplest kernel producing Baxter shifts. thus the whole becomes

$$\frac{2}{1+q^m}E(m) = 2K(p)[mq]_q \tag{6.47}$$

which becomes a Baxter kernel times the quantum number.

in above, the field extension and with quaternion on it naturely give a knots invariants,



6.1 Sharp q-Mellin Transforms and Quantum-Torus Representations

The preceding example suggests that theta products and their root-of-unity specializations may provide a useful language for studying knot-related generating functions. We next examine the sharp q-Mellin transform and its theta-function representation, and investigate the quantum-torus structure suggested by its scaling properties.

For the sharp Q Mellin transformation,

$$\Psi_k^\sharp = (-a\pi)\prod_{j=1}^{\infty}\frac{(1-q^{j+k})(1-q^{j-k-1})}{(1-q^j)^2}\sum_{j=-\infty}^{\infty}q^{(k-j)^2/4} \tag{6.48}$$

rewrite it

$$\Psi_k^\sharp = (q^{k+1};q)_\infty(q^{-k};q)_\infty(q;q)_\infty^{-2}\Theta(k) \tag{6.49}$$

$$\Theta(k) = \sum_{m\in\mathbb{Z}}q^{(m-k)^2/4} \tag{6.50}$$

$$(q^{k+1};q)_\infty = \prod_{m=1}^{\infty}(1-q^{m+k}) \tag{6.51}$$

where $(q^{k+1};q)_\infty$ is a continuous shift of the root lattice. and

$$\Theta(0) = \sum q^{m^2/4} = \vartheta_3(\tau/2) \tag{6.52}$$

$$\Theta(1) = \sum q^{(m-1)^2/4} = \vartheta_4(\tau/2) \tag{6.53}$$

and

$$\sum_j q^{(k-j)^2/4} = q^{k^2/4}\vartheta_3(\frac{k\tau}{2}|\frac{\tau}{2}) \tag{6.54}$$

lift it in two variables use

$$\Theta(z;\tau) = \sum_{n\in\mathbb{Z}} e^{\pi i\tau n^2 + 2\pi i n z} \tag{6.55}$$

then

$$\Psi_k^\sharp = (-\pi a)q^{k^2/4}(q^{1+2z/\tau};q)_\infty(q^{-2z/\tau};q)_\infty(q;q)_\infty^{-2}\vartheta_3(z|\frac{\tau}{2}) \tag{6.56}$$

since

$$\vartheta_3(z+\frac{\tau}{2}|\frac{\tau}{2}) = q^{-1/8}e^{-2\pi i z}\vartheta_4(z|\frac{\tau}{2}) \tag{6.57}$$

thus

$$Y\Psi_k^\sharp = (-\pi a)q^{k^2/4}q^{-1/8}e^{-2\pi i z}\frac{1-q^{\pm 2z/\tau}}{1-q^{\pm 2z/\tau+1}}(q^{1+2z/\tau};q)_\infty(q^{-2z/\tau};q)_\infty(q;q)_\infty^{-2}\vartheta_4(z|\frac{\tau}{2}) \tag{6.58}$$

if acting on Y^2, then theta term becomes

$$e^{-2\pi i z}\vartheta_3(z|\frac{\tau}{2}) \tag{6.59}$$

thus $\Psi_k^\sharp$ is a eigenfunction under Y after normalization.

we normalize it

$$\tilde{\Psi}^\sharp = \frac{\Psi_k^\sharp}{\Theta(k)} = (q^{k+1};q)_\infty(q^{-k};q)_\infty(q;q)_\infty^{-2} \tag{6.60}$$

when we take $k = 1/2$, the factors are

$$(q^{3/2};q)_\infty(q^{-1/2};q)_\infty \tag{6.61}$$

after applying identities such as the Jacobi triple product and the duplication formulas for q-Pochhammer symbols

$$(q^{3/2};q)_\infty(q^{-1/2};q)_\infty = -q^{-1/2}(q;q)_\infty^2\frac{\vartheta_{10}(q^{1/2})}{(q;q)_\infty} \tag{6.62}$$

then

$$\tilde{\Psi}^\sharp = -q^{13/24}\frac{\vartheta_2(\frac{\tau}{2}|\tau)}{\eta(\tau)} \tag{6.63}$$

where

$$\eta(-\frac{1}{\tau}) = \sqrt{-i\tau}\eta(\tau) \tag{6.64}$$

$$\vartheta_2(\frac{\tau}{2}|-\frac{1}{\tau}) = \sqrt{-i\tau}e^{\pi i z^2/\tau}\vartheta_4(\frac{1}{2}|\tau) \tag{6.65}$$

thus under

$$\tilde{\Psi}^\sharp(-\frac{1}{\tau}) = -\exp(-\frac{13\pi i}{12\tau} + \frac{\pi i\tau}{4})\frac{\vartheta_4(\frac{1}{2}|\tau)}{\eta(\tau)} \tag{6.66}$$

and

$$\tilde{\Psi}^\sharp(\tau+1) = -e^{\pi i}\frac{\vartheta_1(\frac{\tau}{2}|\tau)}{\vartheta_2(\frac{\tau}{2}|\tau)}\tilde{\Psi}^\sharp(\tau) \tag{6.67}$$

under S, T, the basis is

$$\{\vartheta_1(\frac{\tau}{2}), \vartheta_2(\frac{\tau}{2}), \vartheta_3(\frac{\tau}{2}), \vartheta_4(\frac{\tau}{2})\} \tag{6.68}$$

for two variables

recall

$$\mathcal{F}(\tau) = \eta(\tau)\frac{\vartheta_4(2\tau)}{\vartheta_3(2\tau)} \tag{6.69}$$

then

$$\mathcal{F}(\tau) = q^{1/24}(q;q)_\infty \frac{(q;q^2)_\infty^2}{(-q;q^2)_\infty^2} \tag{6.70}$$

observe

$$(q;q^2)_\infty = \prod_{m\geq 0}(1-q^{2m+1}) \tag{6.71}$$

only odd roots, and

$$(-q;q^2)_\infty = \prod_{m\geq 0}(1+q^{2m+1}) \tag{6.72}$$

PDE

$$2\pi i q\partial_q \mathcal{F} = i\pi V(\tau)\mathcal{F} \tag{6.73}$$

we do replacement

$$q\partial_q \mapsto \frac{T_q - 1}{\log q} \tag{6.74}$$

where

$$T_q : q \mapsto q^2 \tag{6.75}$$

then

$$T\mathcal{F}(\tau) = exp(\Phi(\tau))\mathcal{F}(\tau) \tag{6.76}$$

and

$$\Phi(\tau) \sim \int_\tau^{2\tau} V(s)ds \tag{6.77}$$

for our case

$$T : \tau \mapsto 2\tau \tag{6.78}$$

$$X : f(\tau) \mapsto e^{2\pi i\tau} f(\tau) \tag{6.79}$$

thus

$$TXT^{-1} = X^2 \tag{6.80}$$

it is a rank-1 quantum torus with doubling automorphism. compare with

$$TXT = X^{-1} \tag{6.81}$$

identify in DAHA give $X^3 = 1$ thus, it is a rank 1 DAHA and the modular $PSL(2,\mathbb{Z})$ at the CM point $x^2 + x + 1 = 0$, more precisely, take DAHA hecke operator $T \to 1$, but keep the modular double torus which

$$\mathcal{A}_{q,\bar{q}} = \langle X, Y | YX = qXY, YX = \bar{q}^{-1}XY, q = e^{2\pi i\tau}, \bar{q} = e^{-2\pi i/\tau}\rangle \tag{6.82}$$

under the modular trans S, T, and under the CM point of $x^3 + x + 1$, $(ST)^3 = 1$, then the q becomes the root of unity, then it becomes a Z3 graded noncommutative torus, under monomial rep

$$deg(XY) = deg(X) + deg(Y)\, mod\, 3 \tag{6.83}$$

which more precisely the semidirect product of a quantum torus with the doubling endomorphism.

$$\mathcal{A} = \mathbb{C}[X^{\pm}] \rtimes \langle T\rangle, TXT^{-1} = X^2 \tag{6.84}$$

If we set

$$R(\tau) = \frac{\vartheta_4(2\tau)}{\vartheta_3(2\tau)} \tag{6.85}$$

then

$$R(2\tau) = \frac{\vartheta_4(4\tau)}{\vartheta_3(4\tau)} = \frac{\vartheta_3(2\tau) - \vartheta_4(2\tau)}{\vartheta_3(2\tau) + \vartheta_4(2\tau)} = \frac{1 - R(\tau)}{1 + R(\tau)} \tag{6.86}$$

6.2 Oscillator Representations and Affine Superalgebras

The appearance of theta and eta functions also motivates an oscillator interpretation of the modular products. We compare the constructions based on $\mathfrak{gl}(1|1)$ and $\mathfrak{osp}(1|2)$, examining how their affine denominator identities lead to different theta-function and mock-theta structures.

The two kinds of modular forms that we find under the force of the 48th root of unity actually could be represented by two kinds of oscillators.

Built $\mathfrak{gl}(1|1)$ for A_1, the Weyl group

$$W = \{1, s\} \tag{6.87}$$

use the Dunkle operator

$$D = \partial_x + k\frac{1 - s}{x} \tag{6.88}$$

the commutator

$$[D, x] = 1 + 2ks \tag{6.89}$$

Now we add a Clifford $cl(1)$ module on it, which

$$\{\psi, \psi\} = 0 \quad \{\psi^\dagger, \psi^\dagger\} = 0 \quad \{\psi, \psi^\dagger\} = 1 \tag{6.90}$$

We define

$$E = x\psi^\dagger \quad E^2 = 0 \quad F = D\psi \quad F^2 = 0 \tag{6.91}$$

the anticommutator

$$\{E, F\} = Dx - (1 + 2ks)\psi^\dagger\psi \tag{6.92}$$

define

$$N_f = \psi^\dagger \psi \quad H = xD \tag{6.93}$$

then

$$\{E, F\} = H + (1 + 2ks)(1 - N_f) \tag{6.94}$$

further set

$$C = 1 + 2ks \tag{6.95}$$

the $\mathfrak{gl}(1|1)$ generated by

$$\mathfrak{gl}(1|1) = \{E, F, N, C\} \quad N = H - (1 + 2ks)N_f \tag{6.96}$$

where

$$[E, F] = H \quad [C, E] = 0 \quad [C, F] = 0 \quad [C, H] = 0 \tag{6.97}$$

For affine $\hat{\mathfrak{gl}}(1|1)$

$$\prod_{n \geq 1} \frac{(1 - q^n)^2}{(1 - zq^n)(1 - z^{-1}q^n)} \tag{6.98}$$

For affine $\hat{\mathfrak{gl}}(1|1)$, denominator identity is

$$\prod_{n \geq 1} \frac{(1 - q^n)^2}{(1 - zq^n)(1 - z^{-1}q^n)} \tag{6.99}$$

$\hat{\mathfrak{gl}}(1|1)$ is controlled by isotropic odd root, leading to Appell-Lerch/mock-theta behavior.

Now we construct $\mathfrak{osp}(2|1)$. We couple a (odd element in a Clifford algebra) Majorana fermion with it (with normalization),

$$\{\psi, \psi\} = 1 \quad \psi^2 = \frac{1}{2} \tag{6.100}$$

set

$$Q_- = D\psi \quad Q_+ = x\psi \tag{6.101}$$

then

$$Q_-^2 = \frac{1}{2}D^2 \quad Q_+^2 = \frac{1}{2}x^2 \quad \{Q_-, Q_-\} = 2F \quad \{Q_+, Q_+\} = 2E \tag{6.102}$$

then

$$\{Q_+, Q_-\} = xD\psi^2 + Dx\psi^2 \tag{6.103}$$

define

$$H = \frac{1}{2}(xD + Dx) \tag{6.104}$$

then

$$\{Q_+, Q_-\} = H \tag{6.105}$$

then

$$[H, Q_-] = -Q_- \quad [H, Q_+] = Q_+ \quad [E, Q_-] = -Q_+ \quad [F, Q_+] = Q_- \tag{6.106}$$

$$\{Q_-, Q_-\} = 2E \quad \{Q_+, Q_+\} = 2F \quad \{Q_+, Q_-\} = H \tag{6.107}$$

they form the algebra $\mathfrak{osp}(2|1)$. For affine $\hat{\mathfrak{osp}}(1|2)$, denominator identity is

$$\prod_{n\geq 1} \frac{(1+zq^{n-\frac{1}{2}})(1+z^{-1}q^{n-\frac{1}{2}})}{(1-zq^n)(1-z^{-1}q^n)} \tag{6.108}$$

$\hat{\mathfrak{osp}}(1|2)$ is controlled by a non-isotropic odd root, leading to the ordinary Jacobi theta system of functions $\vartheta_2, \vartheta_3, \vartheta_4$.

6.3 DAHA Knot Invariants and the BPS Perspective

We now return to DAHA knot invariants and ask whether the preceding analytic and representation-theoretic structures admit a common geometric interpretation. After recalling the algebraic construction of torus-knot invariants, we formulate possible connections with six-dimensional $(2,0)$ theory and BPS states as conjectural directions rather than established equivalences.

Cherednik defines DAHA knot invariants for all reduced root systems; however, a fully developed knot-theoretic interpretation is presently established primarily in type A_1. For other root systems, the construction exists in the form of DAHA–Macdonald-type invariants, but their geometric and topological meaning remains more subtle and is still actively under investigation.

For any reduced root system $\triangle$, , it's Weyl group is W. The Macdonald polynomials satisfy:

$$D_f(q,t)\mathcal{P}_\lambda(x;q,t) = f(t^\rho q^\lambda)P_\lambda(x;q,t) \tag{6.109}$$

where $D_f(q,t)$ are Macdonald operators, ρ is weyl vector, and λ is dominant weight.

For torus knots, Cherednik defines:

$$J^{\triangle}_{\lambda,s}(q,t) = \langle \gamma_{r,s} \cdot \mathcal{P}^{\triangle}_\lambda(x;q,t)\rangle_{eva} \tag{6.110}$$

where $\mathcal{P}^{\triangle}_\lambda(x;q,t)$ is Macdonald polynomial with root system $\triangle$, $\gamma_{r,s} \in Aut(\mathcal{H}_{q,t(\triangle)})$ is DAHA braid element, and $\langle\cdot\rangle_{eva}$ is DAHA evaluation functional.

For $sl(2)$ case,

$$\gamma_{r,s} = X^r Y^s \tag{6.111}$$

For general case, it have link with projective group PSL_2

$$\gamma_{r,s} \in PSL_2(\mathbb{Z}) \ltimes Aut(\mathcal{H}_{q,t(\triangle)}) \tag{6.112}$$

(he has proofed that for any root system, DAHA well defined, Macdonald polynomials exist, evaluation functinal exists, and $J^{\triangle}_{\lambda,s}(q,t)$ well defined algebraically.)

$$Y^{-1}X^{-1}YX = q \tag{6.113}$$

$$J^{A_1}_{\lambda,s}(q,t) \tag{6.114}$$

the torus knot $T(r,s)$ is encoded by $PSL(2,\mathbb{Z})$ by

$$X \mapsto X^r Y^s \tag{6.115}$$

As for $SU(2)$, here we consider a generic q which is not truncated, unless, it's a infinty series, hard to further compare with other things.

$$V_n \cong Sym^{n-1}(\mathbb{C}^2) \tag{6.116}$$

$$\gamma_{2,3} \sim \begin{pmatrix} 2 & * \\ 3 & * \end{pmatrix} \tag{6.117}$$

$$J^{A_1}_{\lambda,s}(q,t) \tag{6.118}$$

We aim to reinterpret DAHA and its associated knot invariants within the framework of 6D $(2,0)$, motivated by several considerations. First, We am influenced by the roles of Ω-deformation in 4D $\mathcal{N}$ introduced by Nikita Nekrasov. In particular, We wonder whether the familiar "double torus" structure (ϵ_2, ϵ_2) originated from geometry of S^3 4-sphere, which is a geometric deformation, which is seen as gluing of two tori, or whether it reflects a more fundamental hidden structure, which it is a algebraic deformation. Also, in known extension to 5D, the deformation parameters naturally appear as $(e^{R\epsilon_1}, e^{-R\epsilon_2})$. The parameter R is from the compactification of 5D $\mathcal{N} = 1$. (The intermediate 5D description, while difficult to completely avoid, will be kept largely offstage in first few sections. Our primary concern in next section is the direct passage from 6D to 4D; the 5D interpretation will reappear at a more appropriate moment.) However, since 4D $\mathcal{N} = 2$ can also arise from 6D $(2,0)$ theory via the class $\mathcal{S}$ construction on punctured surface $\Gamma_{g,n}$, the geometric origin of the relevant circle becomes less transparent - there are simply too many circles and not enough bookkeeping.

This motivates the question: is there a more fundamental structure in 5D or 6D whose projection appears in 4D as the Ω-deformation? In this viewpoint, the deformation parameters may be better understood as shadows as a higher dimensional geometry.

The second motivation comes from the so-called qq-characters introduced by Nikita Nekrasov. Their representation-theoretic structure closely resembles that of q-Whittaker functions. Remarkablt, q-whittaker functions arise as a special degeneration of Macdonald polynomials in the limit $t \to 0$, objects that play a central roles in the theory of DADA. Furthermore, q-Whittaker functions satisfy nonsymmetric q-Toda equations, and somewhere surprisingly, this structure is related to the AGT correspondence. Taken together, these observations suggest a coherent web linking qq-characters, q-deformation, and representation theory, hinting that they may be different shadows of a common underlying framework rather than isolated coincidence.

We now arrive at a rather subtle issue concerning the physical interpretation of knots in the DAHA framework. One cannot simply declare that "the knots live on the D-modules" and regard the problem as solved by terminology alone. Such language risks obscuring a genuine conceptual gap: the underlying structures originate from rather different settings, one essentially algebraic and the other analytic (or, after categorification, homological). Recall, for comparison, the familiar regularization of vacuum energy in string theory. There, divergent mode sums are rendered meaningful through a choice of analytic continuation. Conceptually, this resembles a completion procedure: not unlike p-adic

completion altering convergence properties, although implemented through analytic rather than non-Archimedean methods.

By this point, essentially the only candidate still standing is the BPS condition. Supersymmetry conveniently provides the familiar sequence of $1/2, 1/4, 1/8, \cdots$ BPS sectors almost for free. Encouraged — perhaps dangerously so — by this fact, we venture the following conjecture.

Conjecture 1. The two-variable DAHA knots refuse to live on an ordinary affine plane and instead insist on a (q,t)-dependent 2-adic one.

We recall the BPS mass formula governing stable states with electric and magnetic charges (n_m, n_e) [8]

$$M^2 = 2|Z|^2 = 2|n_e a(u) + n_m a_D(u)|^2 \tag{6.119}$$

This formula comes with a rather unmistakable torus structure built in. On this basis, we are led to the following conjecture.

Conjecture 2. The DAHA unknot $q = t = 1$ correspondence reproduces the BPS mass formula up to an overall normalization factor.

This formal moduli-curves as formal spectra over $\mathbb{Z}_2$, and theta function viewpoint-curves reconstructed from valuation compatible fourier expansion.

7 DAHA degeneration

Having discussed the relation between DAHA, theta kernels, and knot invariants, we now turn to their possible degeneration toward the four-dimensional Ω-background. Rather than assuming a direct equivalence, we focus on the affine lattice, Weyl symmetry, and deformation parameters that may survive this dimensional reduction.

$$\text{DAHA} \;\to\; \text{Trigonometric DAHA} \;\to\; \text{Rational DAHA}$$

Following the previous discussion, the double affine Hecke algebra (DAHA) $\mathcal{H}$ contains two affine lattice directions, namely the weight lattice and its dual, represented by the commuting families (X) and (Y). Let (T) be a maximal torus of the corresponding Lie group. Then (X) and (Y) may be identified with the character and cocharacter lattices, $X = \mathrm{Hom}(T, \mathbb{C}^\times), Y = \mathrm{Hom}(\mathbb{C}^\times, T)$. The Weyl group (W) acts naturally on both lattices, and the DAHA may be viewed as a deformation of the crossed product algebre. $\mathbb{C}[X] \rtimes W \rtimes \mathbb{C}[Y]$. In this sense, DAHA incorporates two independent affine directions associated with the lattices (X) and (Y), together with the Weyl group symmetry. Throughout this paper, we adopt the following parametrization. Select the following parameterization:

$$q = e^{\hbar} \quad t = e^{k\hbar}, \quad \hbar \to 0 \tag{7.1}$$

and recall

$$Y f(X) = f(qX) = f(e^{\hbar} X) = f((1 + \hbar + \frac{\hbar^2}{2} + \cdots)X) \tag{7.2}$$

Make the above equation stable, which $Yf(X) = X$. It means Passing to the spherical subalgebra amounts to projecting onto the W invariant sector. In this setting, the multiplicative X variables are replaced by symmetric polynomials, which may be identified with functions on the Cartan subalgebra. $\mathfrak{h} = \mathrm{Hom}_{\mathbb{C}}(\mathbb{C}, \mathrm{Lie}(T)$ From this perspective, the trigonometric DAHA may be viewed as a quantization of the multiplicative phase space $T \times T^{\vee}$. Under the degeneration from the trigonometric DAHA to the rational DAHA, the multiplicative lattice variables are linearized: the torus directions are replaced by their corresponding Lie algebras. In particular, the dual lattice direction (Y) degenerates to the additive variable

$$\mathfrak{h}^* \simeq \mathrm{Hom}_{\mathbb{C}}(\mathrm{Lie}(T), \mathbb{C}) \tag{7.3}$$

[5] so that the rational DAHA is naturally associated with the cotangent space $T^*\mathfrak{h} \simeq \mathfrak{h} \oplus \mathfrak{h}^*$. Thus the passage from trigonometric to rational DAHA may be interpreted geometrically as replacing the multiplicative torus variables by additive coordinates on the Cartan subalgebra and its dual.

Now we take the (global) projection

$$e_+ = \frac{1}{|W|} \sum_{w \in W} T_w \tag{7.4}$$

in detail

$$e_+ = \frac{1}{\sum_{w \in W} t^{l(w)}} \sum_{w \in W} t^{l(w)} T_w \tag{7.5}$$

where

$$e_+^2 = e_+ \quad T_i e_+ = t^{1/2} e_+ \tag{7.6}$$

for semisimple A_1 case, $W = \{1, s\}$, then

$$e_+ = \frac{1}{1 + t^{1/2}} (1 + t^{1/2} T) \tag{7.7}$$

then give the algebra;

$$e \mathcal{H}^R_{q,t} e \tag{7.8}$$

Recall the Demazure-Lusztig operator

$$T_i = t_i^{1/2} s_i + (t_i^{1/2} - t_i^{-1/2})(X_{\alpha_i} - 1)^{-1}(s_i - 1) \tag{7.9}$$

Consider the operator

$$E_i^+ = \frac{T_i + t_i^{-1/2}}{t_i^{1/2} + t_i^{-1/2}} (X_{\alpha_i} + 1) \tag{7.10}$$

We interpret E_i^+ as a localized projector onto the s_i-invariant sector near the reflection wall $X_{\alpha_i} = -1$. Indeed, recall the Demazure operator

$$\partial_i f = \frac{f - s_i(f)}{1 - X_{-\alpha_i}}, \tag{7.11}$$

for which the operator $1 + \partial_i$ may be regarded as a symmetrization operator, projecting functions toward their s_i-invariant component. The factor $X_{\alpha_i}+1$ localizes the construction to a neighborhood of the wall $X_{\alpha_i} = -1$.

Introducing a local coordinate

$$\varepsilon = X_{\alpha_i} + 1 \tag{7.12}$$

we may write

$$E_i^+ = e_i^+ \varepsilon \tag{7.13}$$

where e_i^+ is regular along the wall. Consequently,

$$(E_i^+)^2 = \varepsilon E_i^+ + O(\varepsilon^2) \tag{7.14}$$

More generally, there exists an integer (n¿0) such that

$$(E_i^+)^2 - E_i^+ \in (X_{\alpha_i} + 1)^n \tag{7.15}$$

Thus, modulo sufficiently high powers of the local parameter ε, the operator E_i^+ behaves as an idempotent. After dividing by ε, one obtains an operator that acts projectively on the normal slice to the reflection wall. This interpretation is closely related to the geometric picture underlying Dunkl operators. Recall that

$$D_z = \frac{\partial}{\partial z} + \sum_{\alpha \in \Delta_+} c_\alpha \frac{\langle \alpha, z \rangle}{x_\alpha}(1 - s_\alpha), \qquad z \in \mathfrak{h}. \tag{7.16}$$

The singular term $X_\alpha^{-1}(1 - s_\alpha)$ measures the failure of invariance across the hyperplane $x_\alpha = 0$, thereby selecting a preferred normal direction to the reflection wall. From this perspective, E_i^+ may be viewed as a projector onto the local branch that crosses the i-th wall continuously, without acquiring a Weyl-group jump. Such operators naturally appear in equivariant K-theory and in the study of stable envelopes.

By fixing the initial and terminal elements in a reduced reflection chain and quotienting by all intermediate decompositions, equivalently by symmetrizing with respect to the Weyl group action, one obtains spherical vectors and hence spherical functions. From the affine viewpoint, each affine root $[\alpha, k]$ determines a family of affine walls

$$\alpha(x) + k = 0, \qquad k \in \mathbb{Z}. \tag{7.17}$$

Macdonald polynomials may be regarded as functions obtained by imposing compatibility conditions across all affine reflection walls simultaneously. Acting on the Gaussian kernel

$$q^{-(b,b)/2}, \qquad b \in B, \tag{7.18}$$

the intertwining operators produce spherical representatives in the polynomial representation. In particular,

$$\mathcal{M}_b, q^{-(b,b)/2} = \prod_{[\alpha,j] \in \lambda'(\pi_b)} \frac{1 - q_\alpha^j t_\alpha X_\alpha(q^{\rho_k})}{1 - q_\alpha^j X_\alpha(q^{\rho_k})}, \widehat{E}_b \tag{7.19}$$

where $\widehat{E}_b$ denotes the normalized nonsymmetric Macdonald polynomial. This formula expresses the passage from localized wall-crossing data to spherical eigenfunctions. [4]

The algebra $\mathcal{H}_{t,c}$ is Morita equivalent to a suitable quotient of a deformed preprojective algebra [5]. More generally, let

$$\mathcal{H}_{t,q}(V, W) \tag{7.20}$$

denote the symplectic reflection algebra associated with a symplectic vector space V and a finite reflection group W. This algebra may be regarded as a quantization of the symplectic quotient $(V \oplus V^*)/W$.

For root systems of type A, the corresponding symplectic reflection algebra arises naturally from quantum Hamiltonian reduction and provides an algebraic realization of the Calogero–Moser system. This should be distinguished from the Drinfeld–Sokolov (DS) reduction. As we shall discuss in the two-dimensional setting, DS reduction leads to the Toda lattice hierarchy, whereas quantum Hamiltonian reduction produces the Calogero–Moser integrable system.

Let $\mathcal{K}[[t]]$ be the formal power series ring and define[5]

$$\mathcal{H}_{\mathcal{K}} = \mathcal{K} \otimes_{\mathbb{C}[[t]]} \mathcal{H}_{q,t}. \tag{7.21}$$

For the spherical subalgebra $e_+\mathcal{H}_{q,t}e_+$, one has a graded $\mathcal{K}$-algebra isomorphism

$$HH^\bullet\left(e_+\mathcal{H}_{\mathcal{K}}e_+\right) \cong \mathcal{K} \otimes_{\mathbb{C}} H^\bullet\left(\mathrm{Spec}(e_+\mathcal{H}_{0,q}e_+)\right) \tag{7.22}$$

which relates the Hochschild cohomology of the quantized algebra to the cohomology of its classical limit.

A particularly illuminating example is provided by the type A_1 case. The classical quotient singularity $\mathbb{C}^2/\mathbb{Z}_2$ admits a minimal symplectic resolution

$$\mathbb{C}^2/\mathbb{Z}_2 \rightsquigarrow T^*\mathbb{P}^1, \tag{7.23}$$

where the exceptional divisor is the projective line $\mathbb{P}^1$. Thus $\mathbb{P}^1$ appears naturally in the geometry of the resolved Calogero–Moser space. More generally, for higher type A_n, the classical limit of the spherical Cherednik algebra is described by Calogero–Moser spaces or related symplectic quotients rather than by $\mathbb{P}^1$ itself.

The appearance of the projective line in the A_1 case provides a useful geometric illustration of the distinction between Drinfeld–Sokolov reduction and quantum Hamiltonian reduction. Recall that the Kleinian singularity

$$\mathbb{C}^2/\mathbb{Z}_2 \tag{7.24}$$

admits a unique minimal resolution

$$\pi : T^*\mathbb{P}^1 \longrightarrow \mathbb{C}^2/\mathbb{Z}_2, \tag{7.25}$$

whose exceptional divisor is the projective line $\mathbb{P}^1$. Thus $\mathbb{P}^1$ may be viewed as the minimal geometric object resolving the singular quotient, while $T^*\mathbb{P}^1$ carries the natural holomorphic symplectic structure relevant for quantization.

From this perspective, Drinfeld–Sokolov reduction and quantum Hamiltonian reduction probe different aspects of the same geometry. The DS reduction removes the cotangent directions and leads to the Toda hierarchy, whose underlying geometric data are governed by the base curve, represented in the A_1 case by the exceptional divisor $\mathbb{P}^1$. In contrast, quantum Hamiltonian reduction retains the full symplectic structure and produces the Calogero–Moser phase space, which is naturally identified with the cotangent bundle $T^*\mathbb{P}^1$ in the A_1 example.

Equivalently, one may regard the Toda system as arising from the geometry of the base ($\mathbb{P}^1$), whereas the Calogero–Moser system is associated with its symplectic thickening ($T^*\mathbb{P}^1$). The latter admits a nontrivial deformation quantization, realized algebraically by the corresponding symplectic reflection algebra or spherical Cherednik algebra.

The deformation theory is governed by Hochschild cohomology. In particular,

$$HH^2\left(e_+\mathcal{H}_{\mathcal{K}}e_+\right) \tag{7.26}$$

controls infinitesimal deformations of the spherical algebra. The parameters q and t exhaust the available deformation directions, and consequently the quantization is rigid once these parameters are fixed. Therefore all higher deformations are already encoded in the Hochschild cohomology classes of the spherical Cherednik algebra.

8 Last degeneration to 4D Ω

The previous discussion concerned the degeneration of the deformation parameters and their possible compatibility with a non-Archimedean completion. We now shift from the parameter space to the observables themselves, examining how the residue factors in Nekrasov's construction relate to the combinatorial structures of Dunkl–Cherednik operators and Macdonald theory.

We introduce the notation $\mathbb{A}_2$ for the rank-two affine lattice, viewed as an additive lattice inside a two-dimensional affine space. In this language, we consider a double extension structure of the form

$$\mathbb{A}_2 \ltimes W \ltimes \mathbb{A}_2 \tag{8.1}$$

where W acts as the reflection symmetry exchanging the two affine directions. From the perspective of the 6D $(0,2)$ theory, compactified with $\Gamma_{g,n}$ could get 4D $\mathcal{N}=2$ theory we discussed here, the Weyl group structure is inherited from the $SU(2)$ (A_1) singularity sector of the 6D moduli space. Since A_1 contains only a single isolated singularity, without entangled or linked singular strata, the corresponding Weyl action reduces to acting on disrecte points (else it would at least acting on the affine dykind diagrams, which may have lines and loops.) Consequently, the affine two dimensional lattice naturly corresponding to the one X and on Y direction in the DAHA construction, rather than the higher-rank coupled configuration. In this sense, the A_1 geometry contributes only a single reflection degree of freedom, and the action space of the singularity sector is effectively discrete. In particular, the Weyl group of $SU(2)$ is $\mathbb{Z}_2$, so that the effective symmetry structure degenerates to

$$\mathbb{A}_2 \ltimes \mathbb{Z}_2 \ltimes \mathbb{A}_2 \tag{8.2}$$

This should be understood as the DAHA shadow of the 6D $(0,2)$ defect geometry, where the two affine lattice directions encode the two canonical shift structures in DAHA: one corresponding to the 6D instantion (or X-type shift), and the other to the effective local Hamiltonian (or Y-type) shift. Equivalently, onemay regard them as the discrete translation direction and the dual differential (Fourier-conjugate) direction. At this level, the construction is already effectively abelianized, corresponding to the passage to a commutative/semi-classical limit of the underlying non-abelian convolution algebra via deformation-thoretic Fourier transform. The remaining $\mathbb{Z}_2$ factor is then identified with the Weyl reflection symmetry associated to the A_1 singularity sector.

In the standard correspondence between DAHA parameters and the Nekrasov Ω-background, one identifies

$$q = e^{\epsilon_1} \quad t = e^{-\epsilon_2} \tag{8.3}$$

We would like to reinterpret the (q,t)-parameter space from the viewpoint of a 2-adic (or more generally, non-archimedean) completion, in order to test whether elliptic deformations preserve the expected BPS filtration of the form $1, 1/2, 1/4, \cdots$. The parameters (ϵ_1, ϵ_2) naturally form an affine lattice in two dimensions. Under a rational slope constraint, we consider the subvarieties

$$q^a t^{2-a} = 1 \quad a \in (0,2) \cap \mathbb{Q} \tag{8.4}$$

Writing $a = \frac{m}{n}$, this becomes

$$\{(q,t) \in (\mathbb{C}^\times)^2; q^m t^{2n-m} = 1\} \tag{8.5}$$

which defines a codimension-one subtorus of $(\mathbb{C}^\times)^2$. Near the identity, we set

$$q = 1 + u \quad t = 1 + v \tag{8.6}$$

so,

$$(1+u)^m (1+v)^{2n-m} = 1 \tag{8.7}$$

Taking logarithms yields

$$m\, log(1+u) + (2n-m)\, log(1+v) = 0 \tag{8.8}$$

Since $\log(1+x)$ is well-defined in the formal (or p-adic) neighborhood of $x = 0$, the leading-order constraint reduces to the linear relation

$$mu + (2n-m)v = 0 \tag{8.9}$$

Choose a unimodular change of basis $(m, 2n-m), (c_1, c_2)$ with determinant ± 1. Introducing new coordinates

$$w = q^m t^{2n-m} \quad s = q^{c_1} t^{c_2} \tag{8.10}$$

the constraint becomes simply $w = 1$. Formally, this induces a projection of completed rings

$$\mathbb{Z}_p[\![u, v]\!] \mapsto \mathbb{Z}_p[\![s-1]\!] \tag{8.11}$$

which remains p-adically complete. Hence, imposing the global constraint $q^m t^{2n-m} = 1$ is compatible with the non-archimedean completion and does not interfere with the expected BPS grading structure.

Finally, under this degeneration—from elliptic DAHA to rational DAHA, and further to the spherical rational limit—the parameters exhibit the expected interchange symmetry

$$q \leftrightarrow t^{-1} \quad \epsilon_1 \leftrightarrow \epsilon_2 \tag{8.12}$$

as anticipated from the underlying Ω-background duality.

9 Partitions , Characters in Dunkel operators

The preceding calculation identifies the residue factors, difference operators, and reflection kernels appearing in the matrix observable, but these local ingredients do not by themselves explain the constraints on the full instanton partition function. We therefore turn to qq-characters, which provide a natural framework for formulating global Ward identities on the sum over instanton fixed points.

Recall the matrix-type observable[7]

$$\Xi_{ij}(x) = Y_{\omega+1}(x + \nu_i + \epsilon)\, \delta_{ij} + \frac{R_\omega(\nu_i + \epsilon_1)}{R'_\omega(\nu_i)} \frac{P_\omega(x + \nu_j)}{Y_\omega(x + \nu_j)} \frac{q_\omega}{\nu_i - \nu_j + \epsilon_1}. \tag{9.1}$$

We first isolate the combinatorial prefactor coming from the residue structure:

$$\frac{R_\omega(\nu_i + \epsilon_1)}{R'_\omega(\nu_i)} = \prod_{k \neq i} \frac{\nu_i - \nu_k + \epsilon_1}{\nu_i - \nu_k}. \tag{9.2}$$

Setting

$$x_i = \nu_i, \qquad \beta = \epsilon_1, \tag{9.3}$$

this becomes the standard deformation of the Vandermonde ratio

$$\Delta_\beta(x) = \prod_{i \neq j} \frac{x_i - x_j + \beta}{x_i - x_j}, \tag{9.4}$$

which is precisely the coefficient appearing in the rational degeneration of Macdonald–Cherednik theory, and simultaneously the Nekrasov instanton measure factor in the Ω-background. Passing to multiplicative coordinates

$$X_i = e^{x_i}, \qquad q = e^{\epsilon_1}, \qquad t = e^{-\epsilon_2}, \tag{9.5}$$

one obtains the trigonometric lift

$$\prod_{i \neq j} \frac{qX_i - X_j}{X_i - X_j}, \tag{9.6}$$

which is exactly the structure coefficient of the first Macdonald difference operator. In DAHA language this corresponds to the spherical Macdonald operator

$$\mathcal{M}_1 = \sum_i \left(\prod_{j \neq i} \frac{tX_i - X_j}{X_i - X_j} \right) T_{q,X_i}, \tag{9.7}$$

with T_{q,X_i} the q-shift operator.

$$T_{q,x_i} f(x_1, \cdots, x_n) = f(x_1, \cdots, qx_i, \ldots, x_n) \tag{9.8}$$

The specialization $t = q^k$ (in particular $k = 1$ in the present normalization) matches the equal-coupling Ω-background limit. Next consider the kernel:

$$\frac{1}{\nu_i - \nu_j + \epsilon_1} = \frac{1}{x_i - x_j + \beta}. \tag{9.9}$$

In the regime $\beta \ll |x_i - x_j|$, we expand

$$\frac{1}{x_i - x_j + \beta} = \frac{1}{x_i - x_j} \left(1 - \frac{\beta}{x_i - x_j} + O(\beta^2)\right). \tag{9.10}$$

At leading order this becomes a reflection-type kernel

$$\frac{1}{x_i - x_j + \beta} \sim \frac{1 - s_{ij}}{x_i - x_j}, \tag{9.11}$$

where s_{ij} denotes the Weyl reflection (transposition) exchanging $x_i \leftrightarrow x_j$.

For the Nekrasov Y-observable we recall

$$Y(x) = \prod_{\square \in \lambda} \frac{x - \phi_\square}{x - \phi_\square - \epsilon}, \tag{9.12}$$

which plays the role of a spectral object dual to Macdonald eigenfunctions $P_\lambda(X; q, t)$. This realizes a spectral (Weyl/bispectral) duality

$$Y(x) \quad \longleftrightarrow \quad P_\lambda(X; q, t). \tag{9.13}$$

Collecting all pieces, the observable $\Xi_{ij}(x)$ can be reorganized as

$$\Xi_{ij}(x) = X^{\vee W}_{i,\lambda+1} + \mathcal{M}_1 \frac{1 - s_{ij}}{x_i - x_j} \frac{P_\lambda(X; q, t)}{X^{\vee W}_{i,\lambda}}. \tag{9.14}$$

Here, $\mathcal{M}_1$ encodes the DAHA/Macdonald difference operator; $(1 - s_{ij})/(x_i - x_j)$ is the rational Dunkl reflection kernel; $P_\lambda(X; q, t)$ is the Macdonald eigenfunction; $X^{\vee W}_{i,\lambda}$ denotes Weyl-dual normalization; $\lambda \mapsto \lambda + 1$ corresponds to adding a box / instanton excitation.

In the DAHA framework, qq-characters can be formulated as non-perturbative Ward identities expressed in terms of spherical Double Affine Hecke Algebra operators acting on the equivariant K-theory of Nakajima quiver varieties. Let $\mathcal{H}_{\mathrm{DAHA}}$ denote the (spherical) DAHA associated to the relevant root system, and let $\mathcal{M}_i$ be the commuting Macdonald difference operators generated by the Dunkl–Cherednik presentation. Then the Nekrasov partition function

$$Z_{\mathrm{inst}}(\mathbf{a}, q, t) = \sum_{\{\lambda\}} \prod_{v \in \mathrm{vertices}} \mathrm{Weight}_v(\lambda) \tag{9.15}$$

is a cyclic vector in a representation of $\mathcal{H}_{\mathrm{DAHA}}$, realized on the fixed-point basis of Young diagrams.

In this setting, a qq-character is an element

$$\chi \in \mathcal{H}^{\mathrm{inv}}_{\mathrm{DAHA}} \tag{9.16}$$

constructed as a screening-invariant combination of Dunkl operators, satisfying

$$[\chi_{qq}, D_\alpha] = 0, \tag{9.17}$$

for all screening currents S_α associated with simple roots α.

Equivalently, χ acts as a family of difference constraints on Z_{inst}:

$$\chi_{qq} Z_{\mathrm{inst}} = \Lambda(\mathbf{a}, q, t)\, Z_{\mathrm{inst}}, \tag{9.18}$$

where $\Lambda(\mathbf{a}, q, t)$ is the corresponding eigenvalue encoding Coulomb and equivariant parameters. From the Macdonald/Cherednik algebra realization, these constraints are generated by the spherical Macdonald operators

$$\mathcal{M}_r = \sum_{|I|=r} \left(\prod_{i\in I,\, j\notin I} \frac{tX_i - X_j}{X_i - X_j} \right) T_{q,I}, \tag{9.19}$$

and the -characters arise as Weyl-invariant combinations of $\mathcal{M}_r$ that remain stable under the full summation over fixed points $\{\lambda\}$.

In particular, the quiver fixed-point sum

$$Z_{\mathrm{inst}} = \sum_{\{\lambda\}} Z_{\{\lambda\}} \tag{9.20}$$

is constrained by the requirement that all $\mathcal{H}_{\mathrm{DAHA}}$-screening charges annihilate the total contribution, ensuring consistency across all BPS sectors. Thus, qq-characters are precisely the DAHA-invariant (screening-closed) difference operators encoding global Ward identities on the equivariant K-theoretic instanton partition function.

10 Discussion

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11 Discussion

The main purpose of this work is to explore whether the algebraic structures of DAHA can provide a common language for understanding the geometry of the six-dimensional $(2, 0)$ theory, its four-dimensional Ω-deformation, and the representation theory associated with ADE singularities. We do not claim that these structures are already equivalent. Rather, we start from the observation that the weight and coweight lattices, the two families of DAHA variables X and Y, and the deformation parameters (q, t) naturally suggest a geometry with two interacting directions. From the six-dimensional viewpoint, we would like to interpret the X-variables as encoding instanton sectors and the Y-variables as describing

the corresponding local Hamiltonians. The resulting picture resembles a generalized state–operator correspondence, in which the instanton sector provides the geometric background for local quantum dynamics.

The theta-function construction provides a second perspective on this problem. We have seen how q-deformed products in the DAHA kernel degenerate to the classical Weyl denominator, while theta-function ratios suggest a relation between modular transformations, divisor geometry, and the curvature of line bundles. This leads us to regard the deformation not merely as an algebraic modification of the Weyl group, but also as a possible geometric deformation of the underlying singularity data. The connection with Hermitian metrics and logarithmic curvature is particularly suggestive, since it gives a way to describe the zeros and poles of the kernel geometrically. However, a complete interpretation requires a careful construction of the relevant line bundles, their automorphy factors, and their curvature forms. We therefore regard the theta-function picture as a geometric framework to be developed further, rather than as a complete identification of DAHA curvature.

A related question is how this structure behaves under degeneration. In the limit $q \to 1$, with $t = q^k$, the q-shift operators approach differential operators, and the Macdonald–Cherednik construction leads toward the rational Dunkl framework. On the gauge-theory side, the residue factors in Nekrasov's observables exhibit Vandermonde-type structures, while the corresponding qq-characters formulate constraints on the full instanton partition function. These observations suggest a possible route from the two-parameter deformation to the differential and combinatorial structures appearing in four-dimensional gauge theory. In particular, the local reflection kernels and the global Ward identities should be understood as different levels of the same problem: the former describe the action near individual configurations, while the latter constrain the sum over instanton fixed points.

The periodic skew Young diagrams provide a useful representation-theoretic model for organizing these configurations. In the singular parameter regime, they encode families of irreducible DAHA modules, while their classical limits can be related to pairs of lattice paths subject to nonintersection conditions. The corresponding determinants and Gaussian binomial coefficients give a concrete way to study the counting problem. From this viewpoint, the passage from ordinary Young diagrams to periodic skew diagrams reflects the additional affine structure, and the q-deformed counting functions retain information that disappears in the classical limit. What remains unclear is the precise physical meaning of this basis: in particular, we have not yet established that its elements are in one-to-one correspondence with BPS states or that the combinatorial counting reproduces the full physical spectrum.

This brings us to the question of the BPS filtration. The parameters $q = e^{\epsilon_1}$ and $t = e^{-\epsilon_2}$ describe the two deformation directions, and rational-slope constraints such as

$$q^m t^{2n-m} = 1$$

select special subvarieties of the parameter space. After passing to a suitable formal neighborhood, such constraints can be studied using completed local rings. This provides a

possible algebraic setting for investigating whether a non-Archimedean completion is compatible with the expected BPS grading. Nevertheless, the formal compatibility of a parameter constraint with completion does not by itself prove that the BPS filtration is preserved. Establishing this requires an explicit relation between the completed algebra, the instanton sectors, and the supersymmetry conditions that distinguish the different BPS fractions.

Finally, we return to the question of instantons in six dimensions. Since the self-dual fields naturally involve the intermediate Jacobian and theta-function structures, it is tempting to look for a description of nonperturbative contributions in terms of trajectories connecting defect or singularity loci. The language of instantons living between singularities is intended as a geometric heuristic, not as a claim that ordinary four-dimensional instantons have a direct six-dimensional counterpart without further construction. Our conjectural picture is that DAHA, theta functions, and periodic diagram combinatorics may provide complementary descriptions of these protected sectors. A more precise formulation should explain how the six-dimensional geometric data determine the two DAHA deformation parameters, how the resulting representation space carries the BPS grading, and whether the limit $q = t = 1$ has a well-defined relation to the Seiberg–Witten central charge and BPS mass formula.

The central question, therefore, is whether the structures identified here are manifestations of a single geometric mechanism or only partially related algebraic analogies. We expect that answering this question will require a construction that keeps track simultaneously of the ADE singularities, the X–Y duality, and the completion of the deformation parameters. For now, DAHA offers a concrete algebraic setting in which to formulate this problem, while the theta-function and periodic-diagram descriptions suggest possible analytic and combinatorial approaches to its resolution. ‴

A DAHA

First, we introduce the affine root system. Let $\triangle$ denote the (non-affine) root system associated with the corresponding Lie algebra, and let W be its Weyl group. We denote $\triangle^+$ for the set of positive roots. With respect to the standard Euclidean inner product on the root space, one obtains the corresponding dual (or coroot) system $\triangle^\vee = \{\alpha^\vee = 2\alpha/\langle\alpha,\alpha\rangle\}$. Further, we have weight lattice

$$\Lambda_{root} = \oplus_{i=1}^n \mathbb{Z}\alpha_i \subset \Lambda_{weight} = \oplus_{i=1}^n \mathbb{Z}\omega_i \tag{A.1}$$

where, ω_is are the fundamental weights defined by $\langle\omega_i, \alpha_i^\vee\rangle = \delta_{ij}$ for simple coroots $\alpha_i^\vee$.

Suppose the original root system has dimension n. We then extend it by one dimension to obtain a space of dimension $n+1$, on which we introduce the associated affine root system. Namely,

$$\tilde{\alpha} = (\alpha, \nu_\alpha j) \in \mathbb{R}^{n+1} \quad \alpha \in \triangle, j \in \mathbb{Z} \tag{A.2}$$

We choose to add a simple root, $\alpha_0 = (-\theta, 1)$. In this new system, the positive roots become

$$\triangle^+ \cup \{[\alpha, \nu_\alpha j], \alpha \in \Delta, j > 0\} \tag{A.3}$$

For convince, we set $\tilde{\alpha}^\vee = \tilde{\alpha}/\nu_\alpha$, thus we have $\alpha_0^\vee = \alpha_0$. Under all settings, the affine Weyl group is $W \ltimes \Lambda_{root}$

We further set $m = 2$ for root systems D_{2k} and C_{2k+1}, $m = 1$ for root systems C_{2k} and B_k, for other cases $m = |\Pi|$. The ring where the algebra live on is $\mathbb{Q}_{\shortparallel,\approx} = \mathbb{Q}[q^{\pm 1/m}, t^{\pm 1/2}]$. We Set

$$t_{\tilde{\alpha}} = t_\alpha = t_{\nu_\alpha}, t_i = t_{\alpha_i} \quad \tilde{\alpha} \in \triangle, 0 \leq j \leq n \tag{A.4}$$

If we restrict to the ADE type case, where the corresponding Dynkin diagrams are simply-laced. As a consequence, there is only a single q and a single t. (which corresponds nicely to the reduction to lower-dimensional cases, Ω deformation only have two parameters, reflected in the corresponding number of parameters.)

Further, given a set of pairwise commutative $\{X_i | i \in 1, \cdots, n\}$s

$$X_{\tilde{b}} = \prod_{i=1}^{n} X_i^{k_i} q^k \quad if\ \tilde{b} = [b, k] \tag{A.5}$$

$$b = \sum_{i=1}^{n} k_i b_i \in B, k \in \frac{1}{2m}\mathbb{Z} \tag{A.6}$$

And with [3]

$$\pi_r(x_\beta) = x_{\omega_r^{-1}(\beta)} \delta^{m(\beta, b_{r^*})} \quad \alpha_{r^*} = \pi_r^{-1}(\alpha_0), r \in O^* \tag{A.7}$$

Definition A.1. [1] The double affine Hecke algebra $\mathcal{H}_{q,t}$ is an algebra generated over the field $C_{q,t}$ by $T_i, 0 \leq i \leq n$, pairwise commutative $X_b, b \in B$ (which satisfy (1.5)), with Weyl group $w \in \Pi$ imposed following constrains.

- (i) $(T_i - t^{1/2})(T_i + t^{-1/2}) = 0 \quad 0 \leq i \leq n$
- (ii) $T_i T_{i+1} T_i = T_{i+1} T_i T_{i+1} \quad i < n-1, j > i+1$
- (iii) $\pi_r T_i = T_j \pi_r$ if $\pi_r(\alpha_i) = \alpha_j$
- (iv) $T_i X_b T_i = X_b X_{a_i}^{-1}$ if $(b, \alpha_i) = 1, 1 \leq i \leq n$
- (v) $T_0 X_b T_0 = X_{s_0(b)} = X_b X_\theta q^{-1}$ if $(b, \theta) = -1$
- (vi) $T_i X_b = X_b T_i$ if $(b, \alpha_i) = 0, 0 \leq i \leq n$
- (vii) $\pi_r X_b \pi_r^{-1} X_{\pi_r(b)} = X_{\pi_r(b)} = X_{\omega_r^{-1}(b)} q^{(b_{r^*}, b)}, r \in O^*$

We prefer the present definition, Although it does not reflect the "double" structure, in which the Y's are introduced subsequently as derived objects. If one were to adopt a formulation treating both X's and Y's as given simultaneously, it may misleadingly suggest that they play symmetric roles. In fact, this is not the case: the Y's are constructed from the X's together with the affine Weyl group action. As we explain in the main text, the Y's should be understood as a family of commuting Hamiltonians.

If we define pairwise commutative elements,

$$Y_i = T_{b_i} \tag{A.8}$$

$$Y_b = \prod_{i=1}^{n} Y_i^{k_i} \quad b = \sum_{1}^{n} k_i b_i \in B \tag{A.9}$$

and satisfied the following relations;

$$T_i^{-1} Y_b T_i^{-1} = Y_b Y_{\alpha_i}^{-1}, \quad (b, \alpha_i) = 1 \tag{A.10}$$

$$T_i Y_b = Y_b T_i, \quad (b, \alpha_i) = 0, 1 \leq i \leq n \tag{A.11}$$

then we extend them to the whole $\mathcal{H}_{q,t}$, Then any given element $H \in \mathfrak{H}$ has unique representation

$$H = \sum_{w \in W} g_w T_w f_w, \quad g_w \in \mathbb{C}_{t,q}[X], f_w \in \mathbb{C}_{t,q}[Y] \tag{A.12}$$

there will be [2]

$$\varepsilon : X_i \to Y_i, Y_i \to X_i, T_i \to T_i^{-1}, t_\nu \to t_\nu^{-1}, q \to q^{-1} \tag{A.13}$$

$$\tau_+ : X_b \to X_b, Y_r \to X_r Y_r q^{-(b_r, b_r)/2}, Y_\theta \to X_0^{-1} T_0^{-2} Y_\theta, T_i \to T_i, t_\nu \to t_\nu, q \to q \tag{A.14}$$

$$Y_0 \to Y_0 T_0^2 X_0 = q^{-1} T_{s_\theta} T_0 X_0 . Y_0 = Y_{\alpha_0} \stackrel{\text{def}}{=} q^{-1} Y_\theta^{-1} \tag{A.15}$$

for

$$1 \leq i \leq n, r \in O^*, X_0 = q X_\theta^{-1} \tag{A.16}$$

For related affine part,

$$\epsilon : T_0 \to (Y_0 T_0 X_0)^{-1} = X_\theta T_{s_\theta}, \pi_r \to X_r T_{u_r^{-1}} \tag{A.17}$$

$$\tau_+ : T_0 \to X_0^{-1} T_0^{-1,}, \pi_r \to q^{(\omega_r, \omega_r)} X_r^{-1} \pi_r = q^{(\omega_r, \omega_r)} \pi_r X_{r^*}^{-1} \tag{A.18}$$

where

$$\pi_r X_{r^*} \pi_r^{-1} = q^{(\omega_r, \omega_r)} X_r^{-1}, X_{r^*} T_{u_r} X_r = T_{u_{r^*}}^{-1} \tag{A.19}$$

In special case, A_1, it is quickly noticed that

$$\varepsilon = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \quad \tau_+ = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \tag{A.20}$$

map (ε, τ) generate the $SL(2, \mathbb{Z})$ S-duality group of the 4d $\mathcal{N} = 4$ SYM/Geometric Langlands duality. From our language, the S duality is a nature character in 6D $(0, 2)$. We would note here, under different singularity pattern in 6D, the underlying S-duality is different in 4D. And they could be extended to a homomorphism of GL_2 up to conjugations by the central elements, the central elements here are formed by T_i.

Under the previous, setting the following automorphisms

$$\tau_- \stackrel{\text{def}}{=} \epsilon \tau_+ \epsilon, \sigma \stackrel{\text{def}}{=} \tau_+ \tau_-^{-1} \tau_+ = \tau_- \tau_+^{-1} \tau_- = \epsilon \sigma^{-1} \epsilon \tag{A.21}$$

these automorphism are unique and preserve $T_1, \cdots, T_n, t, q$

$$\tau_- : Y_b \to Y_b, X_r \to Y_r X_r q^{(\omega_r, \omega_r)/2}, X_0 \to (T_0 X_0 Y_0 T_0)^{-1} = q T_{s_\theta} X_0^{-1} T_0^{-1}, T_0 \to T_0 \tag{A.22}$$

$$\sigma : X_b \to Y_b^1, Y_r \to Y_r^{-1} X_r Y_r q^{(\omega_r, \omega_r)}, Y_0 \to T_{s_\theta}^{-1} X_0^{-1} T_{s_\theta} \tag{A.23}$$

where $b \in P, r \in O'$. Also,

$$\tau_- : T_0 \to T_0, \pi_r \to \pi_r, (r \in O') \tag{A.24}$$

$$\sigma : T_0 \to T_{s_\theta}^{-1} X_\theta^{-1}, \pi_r \to Y_r^{-1} X_r \pi_r q^{-(\omega_r, \omega_r)} = T_{u_r}^{-1} X_{r^*}^{-1} \tag{A.25}$$

In A_1 case, their expression is

$$\tau_- = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \quad \sigma = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \tag{A.26}$$

and they safisfied relation

$$\sigma \tau_+^{-1} \sigma = (\tau_+ \tau_-^{-1} \tau_+) \tau_+^{-1} (\tau_- \tau_+^{-1} \tau_-) = \tau_- \tag{A.27}$$

$$\tau_+ \tau_-^{-1} \tau_+ = \tau_- \tau_+^{-1} \tau_- \tag{A.28}$$

the later one is the braid relation which in group $PSL_2(\mathbb{Z})$. from above, the operator σ^2 commutes with $\tau_\pm$. We should note here that σ^2 general not in DAHA and it is the conjugation by $T_{w_0}^{-2}$ if $w_0 = -1$. And σ^4 is always the conjugation by $T_{w_0}^{-2}$, which is central element in the nonaffine Hecke algebra $H = \langle T_i, 1 \le i \le n \rangle$.

The Demazure-Lusztig operators are

$$\hat{T}_j = t_j^{1/2} s_j + \frac{(t_j^{1/2} - t_j^{-1/2})}{(X_{\alpha_j} - 1)} (s_j - 1),\ 0 \le j \le n \tag{A.29}$$

$$\hat{T}_0 = t_0^{1/2} s_0 + \frac{(t_0^{1/2} - t_0^{-1/2})}{(q X_{\theta_0} - 1)} (s_0 - 1) \tag{A.30}$$

$$\text{where } s_0(X_i) = X_i X_\theta^{-(b_i, \theta)} q^{(b_i, \theta)} \tag{A.31}$$

and they are naturally action on $\mathbb{C}_{q,t}[x]$. under the above assumption, the image of $\hat{H}$ of $H \in \mathfrak{H}$ is uniquely determined from the following

$$\hat{H} f(X) = g(X) \tag{A.32}$$

if

$$Hf(X) - g(X) \in \{ \sum_{i=0}^{n} H_i (T_i - t_i^{1/2}) + \sum_{r \in O'} H_r (\pi_r - 1), H_i, H_r \in \mathfrak{H} \} \tag{A.33}$$

and the aotomorphism τ_- preserves this projection.

When DAHA representated in field $\mathbb{Q}_{t,q}$ of rantional function in $\{X\}$, the X-interwiners are introduced as

$$\Phi_i = T_i + (t_i^{1/2} - t_i^{-1/2})(X_{\alpha_i} - 1)^{-1} \tag{A.34}$$

$$S_i = (\phi_i)^{-1} \Phi_i, G_i = \Phi_i (\phi_i)^{-1}, X_0 = q X_\theta^{-1} \tag{A.35}$$

$$\phi_i = t_i^{1/2} + (t_i^{1/2} - t_i^{-1/2})(X_{\alpha_i} - 1)^{-1} \tag{A.36}$$

where $0 \le i \le n$. They play the roles of proper localization of the affine Hecke algebra $\mathcal{H}_Y$ and satisfy the same relations as the Weyl refections do, which

$$\Phi_w = \Phi_{j_l} \cdots \Phi_{j_1} \quad w = s_{j_l} \cdots s_{j_1} \in W \tag{A.37}$$

Since the the Y are derived from X, thus one can obtains that, for A_n $i > 0$, the Y-interwiners $\epsilon(\Phi_{s_i})$ are the X-interwinders Φ_{s_i} with X_{α_i} replaced by $Y_{\alpha_i}^{-1}$ without changing T and t.

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